Artificial Intelligence and Robotics Center of Excellence

CURRICULUM FOR MASTER OF SCIENCE DEGREE

IN

ARTIFICIAL INTELLIGENCE

October 2024

Addis Ababa, Ethiopia

List of Acronyms

AI: Artificial Intelligence

AII: Artificial Intelligence Institute

AASTU: Addis Ababa Science and Technology University

SE: Software Engineering

PGO: Postgraduate Office

PGC: Postgraduate Program Coordinator

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1. Background of the Program

As the global economy continues to evolve, it influences the advancements in Artificial Intelligence (AI) to transform traditional industries and business models. AI is a form of intelligence; a type of technology and a field of study. AI theory and development of computer systems (both machines and software) are able to perform tasks that normally require human intelligence. AI covers a broad range of domains and applications and is expected to impact every field in the future. Its core idea is building machines and algorithms that are capable of performing computational tasks that would otherwise require human-like brain functions. AI curriculum aims at developing the learner's mindset and skills set towards artificial intelligence and how it is understood and applied. The important principle that artificial intelligence embraces is holistic inclusive and progressive development in immersive ways by problem-solving, creative thinking, and critically analyzing data.

AI has included a mix of symbolic and sub symbolic approaches. The solutions it provides rely on a broad set of general and specialized knowledge representation schemes, problem-solving mechanisms, and optimization techniques. These approaches deal with perception (e.g., speech recognition, natural language understanding, computer vision), problem-solving (e.g., search, planning, optimization), generation (e.g., narrative, conversation, images, models, recommendations), acting (e.g., robotics, task-automation, control), and the architectures needed to support them (e.g., single agents, multi-agent systems). Machine learning may be used within each of these aspects and can even be employed end-to-end across all of them. The study of Artificial Intelligence prepares students to determine when an AI approach is appropriate for a given problem, identify appropriate representations and reasoning mechanisms, implement them, and evaluate them with respect to both performance and their broader societal impact

AI stands at the forefront of innovation, driving significant progress in sectors ranging from healthcare and finance to manufacturing and agriculture. It is a multidisciplinary field that intersects computer science, mathematics, psychology, and more, focusing on the development of intelligent machines that can perform tasks that typically require human intelligence. This includes problem-solving, decision-making, and learning from data. The Master of Science in Artificial Intelligence program is designed to equip students with the principles and knowledge needed to

spearhead advances in AI.

Recent initiatives by Ethiopia, such as the transformation plan and digital Ethiopia, aimed at transitioning the country from an agriculture-based economy to an industrial-led one. Despite these ambitious plans, the adoption of modern, automated systems in industries remains minimal, due to the scarcity of professionals skilled in AI. Recognizing this critical gap, the need for a new MSc program in Artificial Intelligence at Addis Ababa Science and Technology University (AASTU) has become evident. This program aligns with AASTU‘s strategic vision and the national AI policy. This program is proposed by the AI and Robotics Center of Excellence of AASTU. It is specifically designed to address the demand for AI professionals capable of innovating, designing, and implementing intelligent-based systems across various domains such as language, medicine, agriculture, environmental science, etc.

This two-year graduate program aims to produce not only researchers with deep theoretical insights but also practitioners with hands-on expertise in AI. By covering a broad spectrum of AI subfields such as machine learning, natural language processing, speech processing, and computer vision students will gain a comprehensive understanding and practical skills to navigate and contribute to the rapidly evolving landscape of AI technology. Additionally, the curriculum emphasizes ethical considerations, ensuring that graduates are well-equipped to address societal impacts and moral questions arising from AI deployment. This initiative is poised to play a pivotal role in Ethiopia‘s journey towards industrialization and technological advancement, by fostering a generation of AI experts ready to tackle both local and global challenges.

2. Rationale of the Program

Addis Ababa Science and Technology University (AASTU) envisions establishing itself as a hub of technological innovation and research in Ethiopia. In alignment with this mission, the development of a specialized MSc program in Artificial Intelligence (AI) is both timely and essential. The establishment of this program addresses the strategic needs of AASTU's Artificial Intelligence Center of Excellence and is driven by the findings from the needs assessment, which revealed several critical gaps and emerging opportunities in AI education and research.

Alignment with the Thematic Focus of AASTU's AI and Robotics Center

The AI Center at AASTU is committed to tackling national challenges by focusing on thematic areas including Artificial Neural Networks., Evolutionary and Genetic Computing., Computer Vision & Pattern Recognition, Robotics, Industrial Automation and Control, Expert Systems, Speech Processing/Recognition., Natural Language Processing., Machine Learning and IoT. However, the center currently faces a shortage of specialized professionals and researchers capable of leading such initiatives. The MSc program in AI will provide advanced training to students, equipping them with both theoretical and practical expertise in machine learning (ML), deep learning, robotics, and AI ethics—all essential to advancing the center's agenda.

Findings from the Needs Assessment and Industry Demand

The needs assessment survey indicated a growing demand for AI professionals across several industries in Ethiopia, including telecommunications, banking, healthcare, agriculture, and transportation. Key stakeholders, such as Ethio Telecom, the Ministry of Innovation and Technology (MiNT), and Ethiopian Airlines, emphasized the need for talent capable of deploying AI solutions tailored to local needs. Additionally, government agencies like the Information Network Security Agency (INSA) and financial institutions highlighted the importance of AI expertise for cybersecurity and fraud detection.

Addressing Skill Gaps and Research Gaps in Ethiopia

The needs assessment identified critical skill gaps in AI-related fields, including data science, NLP, computer vision, and AI for IoT systems. While there is a growing interest in AI among

undergraduate students, opportunities for postgraduate-level education remain limited in Ethiopia, causing students to seek education abroad. The new MSc program aims to bridge this gap by offering advanced courses, hands-on research opportunities, and collaborations with both local and international partners.

Supporting National AI Strategy and Technological Development Goals

Ethiopia's National Digital Transformation Strategy emphasizes AI and big data as enablers of growth and sustainability. The MSc in AI will play a key role in supporting this national strategy by producing highly skilled graduates capable of driving innovation in both public and private sectors. This aligns with AASTU's strategic vision to lead research efforts that contribute to Ethiopia's sustainable development.

Fostering Collaboration and Innovation Ecosystems

Launching the MSc program will foster collaboration between AASTU, the AI Center, and industrial partners, ensuring that research is problem-driven and aligned with real-world challenges. By creating an ecosystem where academia, industry, and government agencies intersect, the program will not only train the next generation of AI professionals but also catalyze startup creation and innovative AI solutions.

Long-term Impact on Research and Capacity Building

The program will enhance capacity building by producing graduates who can contribute to AI research, policy development, and industry applications. Furthermore, it will serve as a platform to retain local talent and attract international collaborations. With the AI Center at AASTU providing a foundation for interdisciplinary research, the MSc program will ensure that the university remains a leading institution for AI education and research in the region.

In summary, the launch of the MSc in AI program at AASTU will directly respond to the thematic priorities of the university's AI and Robotics Center of Excellence and meet the demands highlighted by the needs assessment survey. The program will serve as a critical pillar in advancing Ethiopia's digital transformation by equipping students with the skills required to innovate and lead in AI.

Program Outcomes

On successful completion of this program the learner will be able to:

- Demonstrate expertise in AI systems, machine learning, optimization techniques, and real-world AI tools and technologies.
- Select and apply advanced AI techniques and tools to improve decision-making.
- Develop, design, and evaluate cutting-edge AI solutions using the latest industry practices.
- Clearly communicate technical AI solutions.
- Evaluate and address ethical issues in the development and deployment of AI systems.
- Conduct independent research in AI.

3. Program Objectives

3.1 General Objective

The general objective of the program is to:

- Produce graduates who will meet both National and International demand in the field of AI.

3.2 Specific Objectives

The program aims to provide students with:

- **Foundational and Advanced AI Knowledge**: The in-depth understanding necessary to navigate the challenges and intricacies of developing and operating advanced AI systems. This encompasses both theoretical knowledge of AI algorithms and practical skills in their implementation.

- **Critical Appraisal and Ethical Consideration:** The ability to critically evaluate a variety of issues related to the AI development lifecycle, emphasizing the professional, legal, and ethical considerations unique to AI technology deployment. This includes understanding the impact of AI on society and the importance of designing AI systems that are fair, transparent, and accountable.

- **Technical Expertise in AI Development**: Skills to conduct comprehensive analysis and

specification, create and evaluate innovative AI design solutions, develop and test AI models, and make informed decisions in the management of AI projects. This objective underscores the importance of both the technical and managerial aspects of AI system development.

- **Interdisciplinary Collaboration**: An understanding that AI development often requires cross-disciplinary knowledge and teamwork. Students will learn to work effectively with professionals from other fields, such as healthcare, finance, and environmental science, to design AI solutions that address complex real-world problems.

- **Lifelong Learning and Ethical Application:** The cultivation of values, attitudes, and competencies that empower graduates to apply the principles and concepts learned in the program to their future careers. This includes a commitment to continuous learning and an ethical approach to the development and deployment of AI technologies.

These objectives place an emphasis on not just the technical competencies required for AI development, but also on the broader implications of AI technologies, including ethical considerations, societal impact, and the necessity for interdisciplinary collaboration.

4. Graduate Profile

The graduation profile of MS students in Artificial Intelligence is designed to produce well-rounded, industry-ready professionals equipped with cognitive abilities, practical skills, and transferable competencies. Through a curriculum that integrates rigorous academic knowledge with hands-on project work, students will gain practical exposure by engaging in internships and partnerships with leading tech companies. This ensures they are well-prepared for industry-driven AI projects, enhancing their employability. Additionally, the inclusion of global case studies and collaborations with international universities will position graduates to compete in the global AI landscape, further enriching their professional and academic experience.

Specifically, the program equips students with the following abilities:

4.1. Cognitive Abilities

Upon completing this program, students should be able to:

- Advocate for a disciplined approach to the development and management of AI systems,

understanding their complexities and the need for structured frameworks.

- Critically assess a variety of AI concepts, tools, and methodologies, understanding their applications and limitations in solving real-world problems.
- Evaluate AI systems against a comprehensive set of criteria including efficiency, accuracy, fairness, and ethical implications.
- Understand and apply the professional, legal, and ethical frameworks crucial to AI development, ensuring responsible use of AI technologies.

4.2. Practical Abilities

Upon completing this program, students should be able to:

- Design and implement solutions to problems using advanced AI and machine learning techniques, tools, and concepts.
- Apply appropriate project management strategies to AI projects.
- Develop AI models using suitable machine learning paradigms.
- Conduct AI projects, applying research methodologies and data analysis techniques to explore AI applications or improve existing ones.

4.3. Transferable Skills

Upon completing this program, students should be able to:

- Learn effectively and independently in a rapidly evolving field for ongoing professional growth in AI.
- Analyze complex problems systematically utilizing AI to devise innovative solutions.
- Communicate AI concepts and their implications clearly and effectively, both in writing and verbally, to a variety of audiences.
- Manage time and resources efficiently in the context of AI projects, demonstrating an ability to prioritize tasks and allocate resources optimally.
- Collaborate effectively in diverse teams with a focus on interdisciplinary cooperation in AI applications.

5. Admission Requirements

An applicant to the MSc in Artificial Intelligence program has to pass an entrance exam and also

fulfill the University's admission requirements. In addition to this, the applicant should be a graduate who has a Bachelor's Degree in Software Engineering, Computer Science, Information Systems, Information Technology, Computer Engineering, and other related fields.

Minimum Admission Requirements for Regular Master's Degree Program

To be considered for enrollment in the Master's program, applicants must:

- Have completed a Bachelor's Degree in either Software Engineering, Computer Science, Information Systems, Information Technology, Computer Engineering, or a related field of study with a minimum CGPA of 2.0 for both Male and Female applicants from a recognized University or Institution, as specified in the respective curriculum.
- Meet the selection criteria, which may include an entrance examination administered by the relevant academic units.

6. Duration of Study

The duration for study for Master's programs shall be as stated as in the university Senate legislation.

7. Teaching and Learning Approach

7.1 Course Work

To graduate, students must successfully complete a list of compulsory and elective courses. These courses are designed to enhance the student's fundamental knowledge and skills in their field of study, enabling them to conduct applied research. Student evaluation will be based on various components, including midterm exams, essays, laboratory sessions, demonstrations, project work, and a final examination.

7.2 Publication of Research Outputs

MSc students are expected to publish an article. The article should be from their research work (Thesis) in a reputable journal or conference proceedings. An article submitted can be considered for fulfilling the program's publication requirement for graduation.

7.3 MSc Thesis

The MSc Thesis proposal must be defended and approved by the examiners. The Thesis must be submitted and defended within the designated study period. It shall demonstrate the candidate's technical expertise in the chosen field and his/her ability to conduct further scholarly work. The examination and final submission procedures of the Thesis shall be based on the guidelines set by the University Senate.

8. Mode of Delivery

The MSc program is currently structured for full-time delivery. However, extension and weekend formats may be introduced to accommodate the growing societal demand for flexible learning options.

9. Assessment and Evaluation Mechanisms

The abilities that graduates of the AI program must acquire are assessed through the following means:

Cognitive Abilities:

- **Theoretical Knowledge**: Assessed through traditional written examinations that include questions on AI theories, algorithms, and data structures.
- **Critical Analysis and Ethical Considerations**: Evaluated through marked essays and case studies that require students to critically appraise AI technologies, including their societal and ethical implications.
- **Problem-Solving with AI**: Project work that involves proposing, designing, and developing AI solutions for real-world problems. This includes computer-based problems that require the application of machine learning models and data analytics.
- **Research and Innovation**: Assessed through dissertation projects or research papers where students explore novel AI concepts or technologies, contributing to the field.

Practical Abilities:

- **Technical Proficiency:** Practical skills are predominantly assessed through individual and group project work that involves developing AI models or systems, demonstrating the application of machine learning, natural language processing, computer vision, and other AI technologies.
- **Project Management and Implementation**: Through the execution of substantial AI projects, including the use of project management tools, version control systems, and collaborative software development practices.

Transferable Skills:

- **Communication**: Directly assessed through presentations and written reports on project work, where students must effectively communicate complex AI concepts and their project findings.
- **Collaboration and Leadership**: Through group projects, assessing the ability to work as part of a team, including roles as team leader, where coordination and leadership skills are critical.
- **Adaptability and Continuous Learning**: While not directly assessed, skills such as the ability to quickly learn and apply new AI tools and technologies are reflected in the quality and innovation of project work and research, enhancing overall student performance.
- **Ethical Application of AI**: Through discussions, essays, and project reflections, assessing

students' understanding and consideration of ethical principles in the development and deployment of AI technologies.

By adjusting the assessment and evaluation mechanisms in this manner, the program ensures that evaluations are comprehensive and aligned with the critical competencies required in the AI field. This not only assesses students' mastery of AI concepts and practical skills but also their ability to apply these skills ethically and effectively. Evaluation will be done based on the midterm exam, essays, laboratory sessions, demonstrations, project work, and final examination, and so on.

10. Grading System

Grading is done as per the University Grading System as follows:

Raw Mark out of 100	Letter Grade	Grade Points
[90,100)	A+	4.00
[85,90)	A	4.00
[80,85)	A-	3.75
[75,80)	B+	3.50
[70,75)	B	3.00
[65,70)	B-	2.75
[60,65)	C+	2.50
[50,60)	C	2.00
<50	F	0.00

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11. Graduation Requirements

The major graduation requirements of the program include:

- Complete all graduate coursework requirements with a minimum CGPA of 3.0;
- a maximum of one "C" or a maximum of two "C+" grades in all courses;

- submit at least one article in peer peer-reviewed journal from the result of the thesis work;
- a minimum of "Satisfactory" rate for Thesis assessment and open defense.

12. Degree Nomenclature

The degree awarded to a student who has completed all the requirements of this program will be entitled as,

In English:

"Master of Science Degree in Artificial Intelligence"

In Amharic**:**

"የሳይንስ ማስተር ዲግሪ በአርቴፊሻል ኢንተሊጀንስ"

13. Staff Profile and Resources

13.1. Staff Profile

No.	Academic Rank	Male	Female	Organization	Subtotal	Remark
1	Associate Professor	3	0	AASTU	2	Dr. Asrat Mulatu Dr. Surafel Lulseged
2	Assistant Professor	11	2	AASTU	12	Dr. Ashenafi Yadessa Dr. Habib Mohammed Dr. Muleneh Mokenen Dr. Kula Kekeba Dr. Hussien Seid Dr. Lemlem Kassa Dr. Girma Nesher Dr. Zeleke Abebaw Dr. Yodit G/Halif Dr. Tilahun Melak Dr. Tesfaye Gidey Dr. Tulu Tilahun Dr. Taye Girma
3	PhD (on study leave)	2	0	AASTU	2	Mr. Biruk Mulatu Mr. Amsalu Samuel
				Total	**17**	

13.2. Resources

In the department, students will be encouraged to actively engage with the material taught, applying

it in innovative ways within the realm of Artificial Intelligence. The curriculum is designed to ground students in both the foundational theories of AI and the practical applications of these concepts in solving real-world problems.

Laboratory Exposure and AI Projects: The understanding of theoretical AI concepts is reinforced through hands-on laboratory work and extensive AI design projects. Students will have the opportunity to work with AI models, data analytics, machine learning algorithms, and more, in a practical setting.

Computer Laboratories: To support the advanced requirements of the AI program, the department will establish two new computer laboratories equipped with the latest Computer Hardware and Software essential for AI development.

Lab I: Dedicated to foundational AI and machine learning courses for 1st year students. This lab will be equipped with software tools for data analysis, machine learning, and simulation environments.

Lab II: Reserved for advanced AI courses for 2nd year students. This lab will focus on deep learning, natural language processing, and robotics, with access to high-performance computing resources and specialized AI development tools.

14. Quality Assurance

To ensure the AI program meets the highest standards of quality and relevance, several quality assurance mechanisms will be implemented:

- **Feedback Mechanisms**: Regular feedback will be sought from students, staff, and stakeholders at the end of each academic semester. This feedback will focus on the curriculum's relevance, the effectiveness of teaching methods, and the adequacy of resources provided.
- **Program Assessment Tools:** The Department Graduate Committee (DGC) will develop and utilize a range of assessment tools and procedures specifically designed to evaluate the quality and standards of the AI curriculum. These tools will consider the rapidly evolving nature of the AI field, ensuring the program remains up-to-date and responsive to both

technological advancements and market needs.

- **Ethical and Societal Considerations**: Part of the quality assurance process will involve evaluating how well the program prepares students to address the ethical and societal implications of AI technologies. This includes fostering a deep understanding of ethical AI use, data privacy, and the potential societal impacts of AI solutions.
- **Interdisciplinary Collaboration:** The quality assurance process will also assess the program's effectiveness in promoting interdisciplinary learning, reflecting the cross-disciplinary nature of AI applications in areas such as healthcare, finance, environmental science, and more.

By focusing on these adapted resources and quality assurance measures, the AI program aims to provide students with a robust education that prepares them for the complexities and opportunities of working in the field of artificial intelligence.

15. Course Coding

The course code is formatted as "**ARIN** abcd". Here, "Ar" stands for "Artificial", and "In" stands for "Intelligence". The "abcd" part consists of digits with the following meaning:

- The first digit, '**a**' indicates the year the course is offered (6 for 1st year).
- The second and third digits, '**bc**' represent the semester (odd numbers for the 1st semester and even numbers for the 2nd semester).
- The last digit, '**d**' denotes the sequence of the course.

16. List of Course

16.1. List of Core Courses

The following table shows the list of the major courses' course codes, course titles, credit hours, lecture hours, and tutorial/ lab hours.

№	Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
1	ARIN6011	Artificial Intelligence	3	2	3
2	ARIN6012	Machine Learning	3	2	3
3	ARIN6013	Natural Language Processing	3	2	3
4	ARIN6014	Research Methods and Seminar	3	3	0
5	ARIN6021	Computer Vision	3	2	3
6	ARIN6022	Deep Learning	3	2	3
7	ARIN6023	Probabilistic Graphical Models	3	2	3
8	ARIN601X	Elective I	3	2	3
9	ARIN701X	Elective II	3	2	3
10	ARIN7022	Master's Thesis	6	0	

16.2. List of Elective Courses

Depending on various factors a batch may select among the following elective courses.

Elective I

№	Course Code	Courses	Cr. Hrs.	Lec.	Tut/ Lab
1	ARIN6024	Cognitive Modeling	3	2	3
2	ARIN6025	Emerging AI Technologies	3	2	3
3	ARIN6026	Speech Processing	3	2	3
4	ARIN6027	Explainable AI	3	2	3
5	ARIN6028	Intelligent Robotics	3	2	3

Elective II

№	Course Code	Courses	Cr. Hrs.	Lec.	Tut/ Lab
1	ARIN7012	Reinforcement Learning	3	2	3
2	ARIN7013	Neurosymbolic AI	3	2	3
3	ARIN7014	Generative AI	3	2	3
4	ARIN7016	AI Ethics and Society	3	2	3
5	ARIN7017	Intelligent Search and Games	3	2	3

17. Semester Course Breakdown

Year I Semester I

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN6011	Artificial Intelligence	3	2	3
ARIN6012	Machine Learning	3	2	3
ARIN6013	Natural Language Processing	3	2	3
ARIN6014	Research Methods and Seminar	3	3	0
Total Hours		**12**	**9**	**9**

Year I Semester II

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN6021	Computer Vision	3	2	3
ARIN6022	Deep Learning	3	2	3
ARIN6023	Probabilistic Graphical Models	3	2	3
ARIN602X	Elective I	3	0	3
Total Hours		**12**	**6**	**12**

Year II Semester I

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN701X	Elective II	3	2	3
ARIN7011	Master's Thesis	6		
Total Hours		**9**		

Year II Semester II

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN7011	Master 's Thesis	6		
Total Hours		**6**		

18. Course Breakdown for Continuing Education Program (CEP)

The semester-based course breakdown of the MSc. degree program in data science for **CEP or weekend program in three years (***i.e. 3 semesters and 1 summers of course work and 3 semesters and 1 summer of thesis work***)** is presented according to the following Tables.

Year I Semester I

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN6011	Artificial Intelligence	3	2	3
ARIN6012	Machine Learning	3	2	3
ARIN6013	Natural Language Processing	3	2	3
Total Hours		**9**	**6**	**9**

Year I Semester II

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN6021	Computer Vision	3	2	3
ARIN6022	Deep Learning	3	2	3
ARIN6023	Probabilistic Graphical Models	3	2	3
Total Hours		**9**	**6**	**9**

Year I Summer

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN6014	Research Methods and Seminar	3	3	0
Total Hours		**3**	**3**	

Year II Semester I

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN602X	Elective I	3	2	3
ARIN701X	Elective II	3	2	3
Total Hours		**6**	**4**	**6**

Year II Semester II - Year III Semester II

Course Code	Courses	Cr. Hrs.	Lec	Tut/ Lab
ARIN7011	Master ‘s Thesis (*)	6	-	-
Total Hours		**6**	-	-

(*)*Note that for MSc Programs Thesis work starts from Year II Semester till graduation/termination semester (i.e. Year III Semester II).*

19. Course Plan

<table>
<tr><td colspan="5">Addis Ababa Science and Technology University
Artificial Intelligence and Robotics Center of Excellence</td></tr>
<tr><td rowspan="2">Course Code</td><td rowspan="2">Course Title</td><td rowspan="2">Cr. Hr.</td><td colspan="2">Hour</td></tr>
<tr><td>Lect.</td><td>Tut./Lab</td></tr>
<tr><td>ARIN6011</td><td>Artificial Intelligence</td><td>3</td><td>2</td><td>3</td></tr>
<tr><td colspan="2">Class year/Semester</td><td colspan="3">Year I/Semester I</td></tr>
<tr><td colspan="2">Pre-requisites course(s)</td><td colspan="3">None</td></tr>
<tr><td colspan="2">Course Owner</td><td colspan="3"></td></tr>
<tr><td colspan="2">Course Rationale</td><td colspan="3">Delivering Artificial Intelligence course at the master's level is essential due to the high industry demand for AI skills and the need for a comprehensive, interdisciplinary education that combines computer science, mathematics, statistics, and cognitive science. This course provides foundational knowledge in core AI concepts like search algorithms, reasoning, and machine learning, while also fostering algorithmic thinking and ethical awareness. It includes practical projects and familiarity with industry tools, enhancing students' hands-on experience and employability. Additionally, it prepares students for advanced research, future technological advancements, and leadership roles, aligning with international educational standards and promoting global competitiveness.</td></tr>
<tr><td colspan="2">Course objectives and competencies to be acquired</td><td colspan="3">Course Objectives
On successful completion of this course the students will be able to:
• Demonstrate critical understanding of the theory and concepts underpinning Artificial Intelligence.
• Critically analyze the capabilities and limitations of current AI techniques.
• Evaluate and apply the principles, models and algorithms of AI to recognize, model and solve various problems.
• Review, compare, and contrast the latest AI industry and research developments.
• Develop and enhance interpersonal communication skills to become a successful member of a working team.</td></tr>
</table>

Course description/content	**Course Description**: The course offers a comprehensive exploration of the foundational theories and concepts of AI. Students will gain a critical understanding of the capabilities and limitations of current AI techniques and learn to evaluate and apply various principles, models, and algorithms to solve complex problems. The course will also cover the latest industry and research developments in AI, encouraging students to review, compare, and contrast these advancements. Additionally, the course aims to develop and enhance students' interpersonal communication skills, preparing them to become effective members of collaborative working teams. Through a combination of theoretical study and practical application, students will be equipped with the knowledge and skills necessary to navigate and contribute to the evolving field of artificial intelligence. **Course content:** The course content includes the following: **Introduction to Artificial Intelligence** ● Defining Artificial Intelligence ● Foundations of AI: Philosophy, Maths, Psychology, Computing, Linguistics, Logic, Probability Theory; Historical Evolution of the field. **Types of AI** ● Weak vs. strong AI; Narrow vs. general vs. super intelligence; ● Classical vs. statistical AI; ● Use cases and examples of applications for the different types of AI. **Business applications of AI** ● The opportunities and challenges of AI for business; ● Building safe, ethical, explainable and responsible AI applications; ● Use cases and examples of AI applications in business. **Agents** ● Precepts, actions, goals, environment; ● Rational agents; Environments representation; ● Agent functions and programs; ● Types of agents: simple reflex agents, reflex agents with state, goal-based agents, utility-based agents. **Solving Problems by Searching** ● Defining search problems; ● Searching for solutions; Performance; ● Uninformed search strategies; Informed Search strategies; Heuristic functions. ● Local search algorithms and optimization problems; ● Adversarial search. **Reasoning** ● Propositional Logic ● First Order Logic

	● Inference in First-Order Logic ● Planning. **Knowledge Representation** ● Ontological Engineering ● Categories and objects ● Semantic networks **Bayesian Networks** ● Quantifying uncertainty ● Bayes‘ rules and its use ● The semantics of Bayesian networks ● Conditional distributions and efficient representation of them ● Inference in Bayesian networks. **Learning** ● Types of learning (e.g., deductive vs. inductive, supervised vs. unsupervised vs. semi-supervised, reinforcement learning) ● Learning from examples (e.g., regression, classification) ● Brief overview of artificial neural networks ● Learning probabilistic models. **Recommended Systems** ● Recommender systems types and models (e.g., collaborative filtering, content-based filtering, multi-criteria systems, etc.) ● Recommender systems challenges (e.g., scalability, sparsity, cold start, etc.)
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	Stuart Russell, Peter Norvig. (2020), Artificial Intelligence, 4. Pearson Higher Education, p.1136, [ISBN: 978-0134610993]. Andrew Burgess. (2018), The Executive Guide to Artificial Intelligence: How to Identify and Implement Applications for AI in Your Organization, 1. Palgrave Macmillan, p.181, [ISBN: 978-3319638195]. Steven Finlay. (2021), Artificial Intelligence and Machine Learning for Business: A No-Nonsense Guide to Data Driven Technologies, 4. Relativistic, p.226, [ISBN: 978-1999325381]. Mariya Yao,Adelyn Zhou,Marlene Jia. (2018), Applied Artificial Intelligence: A Handbook For Business Leaders, TOPBOTS, p.246, [ISBN: 978-0998289021]. ***Other Resources*** Exercises for Artificial Intelligence: A Modern Approach, https://aimacode.github.io/aima-exercis es/

	Code for Artificial Intelligence: A Modern Approach, https://github.com/aimacode/aima-pytho Zhang, Q., Lu, J. & Jin, Y.. (2021), Artificial intelligence in recommender systems, Complex Intelligent Systems, 7, Springer, https://doi.org/10.1007/s40747-020-00212 -w

Addis Ababa Science and Technology University Artificial Intelligence and Robotics Center of Excellence				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6012	**Machine Learning**	3	2	3

Class year/Semester	Year I/Semester I
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationale**: This course is designed to equip students with the essential skills and knowledge required to excel in the rapidly evolving field of machine learning. Given the increasing importance of data-driven decision-making in various industries, proficiency in machine learning methodologies is crucial. The course emphasizes hands-on experience and practical applications, ensuring that students can effectively apply their learning to real-world scenarios. Additionally, by focusing on advanced techniques and industry best practices, the course prepares students to stay at the forefront of technological advancements. Overall, this course aims to foster a deep understanding of machine learning, enabling students to become proficient practitioners and innovators in their respective fields.
Course objectives and competencies to be acquired	**Course objectives** On successful completion of this course the learner will be able to: Select, apply, and evaluate machine learning methodologies to facilitate pre-processing and transformation approaches. Select, formulate, design, implement, and evaluate machine learning algorithms for solving real- world problems using the latest industry practices and standards. Contextualize, investigate, evaluate, and communicate key concepts and advanced techniques for machine and deep learning algorithms. Demonstrate expert knowledge of machine learning algorithms, tools, techniques utilized in real world contexts.
Course description/content	**Course description**: The "Machine Learning" course provides a comprehensive introduction to the fundamental principles and techniques of machine learning. Students will learn to select and apply various machine learning methodologies for data pre-processing, transformation, and analysis. The course covers the design and implementation of machine learning algorithms, emphasizing real-world applications and industry standards. Advanced topics in machine and deep learning will be explored, enabling students to contextualize and communicate complex concepts effectively. By the end of the course, students will possess expert knowledge of machine learning algorithms, tools, and techniques, and will be prepared to tackle real-world problems

	with confidence. **Course Content** **Advanced Regression Models** ● Regression revision; Best-practice for evaluating performance and analyzing for bias and variance, regularization and penalized models, Generalized Linear Modeling **Advanced Classification Models** ● Logistic regression; Naïve-Bayes Classification and Linear discriminant analysis **Ensembles** ● Neural networks, classic topologies, and activation functions. Forward- and back-propagation. Optimization algorithms: gradient descent and stochastic gradient descent. Key parameters for neural networks. Multi-layer perceptions. **Deep Neural Networks I** ● Neural networks, classic topologies, and activation functions. Forward- and back-propagation. Optimization algorithms: gradient descent and stochastic gradient descent. Key parameters for neural networks. Multi-layer perceptrons. ● Initialization, L2 and dropout regularisation, gradient checking and batch and layer normalization; convergence algorithms, learning rate scheduling, Hyperparameter tuning. **Convolutional neural network** ● Overview of convolutional neural networks (CNN). Methodology for stacking layers in a deep network to address multi-class image classification problems. Object detection and the YOLO algorithm. Deep residual learning for image recognition. **Recurrent Neural Network** ● The basic recurrent unit (Elman unit) and LSTM (long short-term memory) unit. Overview of the GRU (gated recurrent unit). Build and train recurrent neural networks. Approaches for mitigating the vanishing gradient problem. **Transformer** ● Encoder-decoder Architecture, attention mechanisms, Position embedding, multi-head attention and self-attention layers, pre-trained language models (e.g., BERT) **Autoencoders and Deep Generative Models** ● Auto-encoders (AEs), Undercomplete, Regularised, Sparse autoencoders; Denoising autoencoders; Variational Autoencoder,

	Generative adversarial networks (GANs), Min-max cost, Conditional GAN **Active learning (AL); Transfer learning (TL); Practical applications of AL and TL** ● Sampling techniques for stream-based and pool-based active learning; transfer learning strategies, multi-task and zero-shot learning, pre-trained models; **Application of Deep Learning Models** ● Selected topics from the following areas will be covered, especially on practical applications: computer vision, natural language processing
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	Ian Goodfellow,Yoshua Bengio,Aaron Courville. (2016), Deep Learning, MIT Press, p.800, [ISBN: 978- 0262035613]. Hastie, T., Tibshirani, R., & Friedman, J.. (2016), The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd. Springer, [ISBN: 978-0387848570]. Kevin P. Murphy. (2012), Machine Learning, MIT Press, p.1104, [ISBN: 978-0262018029]. Gareth James,Daniela Witten,Trevor Hastie,Robert Tibshirani. (2014), An Introduction to Statistical Learning, Springer, p.426, [ISBN: 978-1461471370]. **Other Resources** [Website], Machine Learning Stanford, https://www.coursera.org/course/ml [Website], DataCamp, http://www.datacamp.com [Website], UCI Repository, http://www.ics.uci.edu/~mlearn/MLReposit ory.html [Website], WEKA, http://www.cs.waikato.ac.nz/ml/weka/ [Website], Kaggle Competitions, https://www.kaggle.com/competitions

Addis Ababa Science and Technology University Artificial Intelligence and Robotics Center of Excellence				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6013	**Natural Language Processing**	3	2	3

Class year/Semester	Year II/Semester I
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationale**: NLP plays a crucial role in advancing technologies that interact with and understand human language, such as virtual assistants, language translation services, and automated content analysis tools. Mastering NLP equips students with skills in data processing, machine learning, and linguistics, making them proficient in developing intelligent systems that can process and generate human-like text. This knowledge not only enhances their technical capabilities but also prepares them for careers in fields requiring advanced AI applications, including healthcare, finance, and information technology, thereby ensuring they remain competitive in the evolving job market.
Course objectives and competencies to be acquired	**Course objectives:** Upon successful completion of the course he student will be able to: ● Understand the basic principles and challenges of natural language processing. ● Explore the core NLP tasks, such as text preprocessing, language modeling, named entity recognition, and sentiment analysis. ● Gain hands-on experience in implementing and applying NLP techniques using popular libraries and frameworks. ● Analyze and critically evaluate the performance and limitations of NLP systems. ● Understand the ethical considerations and societal implications of NLP technology.
Course description/content	**Course Description:** The intent of the course is to present a fairly broad graduate-level introduction to Natural Language Processing (NLP, a.k.a. computational linguistics), the study of computing systems that can process, understand, or communicate in human language. The primary focus of the course will be on understanding various NLP tasks as listed on the course syllabus, algorithms for effectively solving these problems, and methods for evaluating their performance. There will be a focus on statistical and neural-network learning algorithms that train on (annotated) text corpora to automatically acquire the knowledge needed to perform the task.

	Course Content: **1. Introduction to Natural Language Processing** ● Definition and Historical development of NLP ● Challenges in Natural Language understanding ● Applications of NLP in various domains **2. Text Preprocessing and Normalization** ● Tokenization, stemming, and lemmatization ● Stop word removal and case normalization ● Text cleaning and data augmentation **3. Language Modeling** ● N-gram models and statistical language models ● Word embeddings and distributed representations ● Contextual language models (e.g., BERT, GPT) **4. Named Entity Recognition** ● Rule-based and machine learning approaches ● Feature engineering and sequence labeling models ● Evaluation metrics and performance analysis **5. Text Classification and Sentiment Analysis** ● Supervised and unsupervised classification methods ● Feature engineering and representation learning ● Sentiment lexicons and dictionary-based approaches **6. Information Extraction and Relation Extraction** ● Extracting structured information from text ● Rule-based and statistical relation extraction ● Knowledge graph construction and reasoning **7. Question Answering and Dialogue Systems** ● Retrieval-based and generative QA models ● Conversational agents and Chabot architectures ● Human-computer interaction in dialogue systems **8. Machine Translation and Text Generation** ● Statistical and neural machine translation ● Text generation models and creative writing assistants ● Evaluation and quality assessment of generated text **9. Ethical Considerations in NLP** ● Bias, fairness, and accountability in NLP systems ● Privacy, security, and user data protection ● Societal impacts and responsible development of NLP
Assessment/Evaluation modalities	Lectures, discussion, presentation, articles review, assignments, lab activities, individual/group project

Attendance requirements	Regular attendance is required
Textbooks and references	Daniel Jurafsky and James H. Martin. "Speech and language processing an introduction to natural language processing, computational linguistics, and speech." Pearson, London, 2000 Chris Manning and Hinrich Schütze, Foundations of Statistical Natural Language Processing, MIT Press. Cambridge, MA. 1999 Goldberg, Yoav. Neural network methods for natural language processing. Springer Nature, 2022.

Addis Ababa Science and Technology University **Artificial Intelligence and Robotics Center of Excellence**				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6014	**Research Methods and Seminar**	3	2	0
Class year/Semester	Year I/Semester I			
Pre-requisites course(s)	None			
Course Owner				
Course Rationale	**Course Rationale**: The rationale behind this course is to equip students with a comprehensive understanding of research methodology, a critical skill set essential for conducting rigorous and impactful research across various fields. In today's data-driven world, the ability to design robust research strategies, collect and analyze data accurately, and interpret results effectively is paramount. This course provides students with the foundational knowledge and practical skills necessary to navigate the complexities of research. By covering a wide range of topics, from defining research problems and designing sampling methods to conducting hypothesis tests and utilizing multivariate analysis techniques, the course prepares			

	students to tackle real-world research challenges. Additionally, the emphasis on report writing and the role of computers in research ensures that students are well-prepared to communicate their findings and leverage technology in their research endeavors. This holistic approach not only enhances students' research capabilities but also fosters critical thinking, analytical skills, and the ability to contribute meaningfully to their respective fields.
Course objectives and competencies to be acquired	**Course Objectives** Upon the successful completion of the course, students will be able to: ● Understand and apply the principles and techniques of research methodology to define and address research problems effectively. ● Develop skills in designing research strategies, including sampling design and measurement and scaling techniques. ● Gain proficiency in various methods of data collection and the subsequent processing and analysis of collected data. ● Master the fundamentals of hypothesis testing, including both parametric and nonparametric tests, and perform Chi-square tests. ● Acquire knowledge of multivariate analysis techniques and their application in research. ● Learn to interpret research findings accurately, write comprehensive reports, and utilize computer tools to enhance research processes.
Course description/content	**Course Description:** This course offers an in-depth exploration of research methodology, covering a broad range of essential topics. Students will begin with an introduction to research methodology, learning to define research problems and design effective research strategies. The course delves into sampling design, measurement, and scaling techniques, as well as various methods of data collection. Students will gain skills in processing and analyzing data, understanding sampling fundamentals, and conducting parametric (standard) hypothesis testing and Chi-square tests. The curriculum also includes nonparametric (distribution-free) tests and multivariate analysis techniques. Additionally, the course emphasizes the interpretation of data and report writing, highlighting the critical role of computers in research. This comprehensive approach equips students with the knowledge and skills necessary to conduct rigorous and impactful research. **Course Content:** **Introduction to Research Methodology** ● Definition and Importance of Research ● Types of Research ● Research Process Overview ● Ethical Considerations in Research **Defining the Research Problem** ● Identifying and Formulating Research Problems

	● Literature Review: Purpose and Process ● Research Questions and Hypotheses ● Theoretical Frameworks in Software Engineering Research **Research Design and Planning** ● Research Design: Definition and Types ● Experimental vs. Non-experimental Research ● Case Studies, Surveys, and Action Research ● Planning and Managing Research Projects **Sampling Design** ● Principles of Sampling ● Probability and Non-probability Sampling Methods ● Sample Size Determination ● Sampling Errors and Bias **Measurement** and Scaling Techniques ● Concepts of Measurement in Research ● Reliability and Validity of Measurement Instruments ● Scaling Techniques: Nominal, Ordinal, Interval, and Ratio Scales ● Developing and Testing Research Instruments **Data Collection Methods** ● Primary vs. Secondary Data ● Data Collection Techniques: Interviews, Questionnaires, Observations, and Document Analysis ● Ensuring Data Quality and Integrity ● Tools and Software for Data Collection **Data Processing and Analysis** ● Data Preparation: Coding, Editing, and Cleaning ● Descriptive Statistics: Measures of Central Tendency and Variability ● Inferential Statistics: Hypothesis Testing, Confidence Intervals, and p-Values ● Software Tools for Data Analysis (e.g., SPSS, R, Python) ● Sampling Fundamentals and Hypothesis Testing ● Basic Concepts of Probability, Sampling Distributions ● Hypothesis Testing: Parametric Tests (t-tests, ANOVA) ● Chi-square Tests for Categorical Data **Advanced Statistical Methods** ● Nonparametric Tests: Mann-Whitney, Wilcoxon, Kruskal-Wallis ● Introduction to Multivariate Analysis Techniques: Factor Analysis, Cluster Analysis, Regression Analysis ● Software Tools for Advanced Statistical Analysis ● Multivariate Analysis Techniques ● Multivariate Analysis of Variance (MANOVA) ● Principal Component Analysis (PCA) ● Structural Equation Modeling (SEM)

	● Applications of Multivariate Techniques in Software Engineering **Interpretation and Report Writing** ● Interpreting Statistical Results ● Writing Research Reports and Papers ● Structuring Research Papers: Introduction, Methods, Results, Discussion ● Best Practices for Publishing in Academic Journals **The Role of Computers in Research** ● Software Tools for Research Management (e.g., EndNote, Mendeley) ● Using Databases and Online Resources for Literature Review ● Automating Data Collection and Analysis ● Collaborative Tools for Research Projects
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	1. Kothari, Chakravanti Rajagopalachari. Research methodology: Methods and techniques. New Age International, 2004. 2. Research Design: Qualitative, Quantitative, and Mixed Methods Approaches, 4th Edition, by John W. Creswell 3. Qualitative Research: A Guide to Design and Implementation 4th Edition, by Sharan B. Merriam (Author), Elizabeth J. Tisdell (Author) 4. The Research Methods Knowledge Base, 3rd Edition, by William M. K. Trochim (Author), James P. Donnelly

Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6021	**Computer Vision**	3	2	3

Class year/Semester	Year I/Semester II
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationale**: The course addresses the increasing demand for technologies capable of extracting meaningful information from visual data. As computer vision applications proliferate across industries such as healthcare, automotive, and entertainment, understanding the core principles and techniques is essential. This course equips students with the knowledge and skills to develop and apply computer vision algorithms for tasks ranging from basic image processing to complex visual recognition and motion analysis, prepARINg them for careers in research, development, and innovation in this rapidly evolving field.
Course objectives and competencies to be acquired	**Course objectives** Upon successful completion of the course the learner will be able to: ● Understand the fundamental principles of image formation, cameras, and color perception essential for computer vision applications. ● Explore techniques for image enhancement and transformation using filters, templates, and image pyramids. ● Learn methods for detecting interest points and extracting local image features crucial for pattern recognition and image understanding. ● Master edge detection algorithms and techniques for fitting lines to extracted edges, facilitating precise image segmentation and analysis. ● Gain insights into machine learning approaches including clustering and classification, and their application to computer vision tasks such as instance recognition and object detection. ● Develop proficiency in high-level vision tasks such as object detection with sliding windows, motion analysis, human action recognition, and image retrieval. ● Explore advanced applications of computer vision in fields such as augmented reality, autonomous systems, medical imaging, and surveillance.
Course description/content	**Course Description**: This undergraduate-level course in Computer Vision introduces fundamental concepts and methodologies aimed at enabling machines to interpret and analyze visual inputs. Beginning with low-level aspects of image perception, including image formation, cameras, and color, the course progresses to mid-level vision topics such as interest point detection and local feature extraction. It also covers high-level vision tasks including face detection/recognition, object recognition, and human motion analysis, providing a comprehensive foundation in computer vision principles. **Course Content**

	1: **Introduction to Computer Vision** ● Overview of computer vision fundamentals and its applications across various domains. 2**. Image Formation, Camera, and Color** ● Understanding how images are formed, camera models, and the basics of color representation in computer vision. 3. **Filters, Templates, and Image Pyramids** ● Techniques for image enhancement, transformation, and hierarchical representation using pyramids. 4. **Interest Points and Image Features** ● Methods for detecting interest points and extracting local features crucial for image understanding and recognition. 5. **Edge Detection and Line Fitting** ● Algorithms for detecting edges in images and fitting lines to extracted edge features. 6. **Machine Learning Overview for Computer Vision** ● Introduction to clustering and classification techniques and their application to computer vision tasks. 7. **Instance Recognition and Bag-of-Words Models** ● Methods for instance recognition and feature-based models like bag-of-words for image understanding. ● Object Detection with Sliding Windows ● Techniques for detecting objects in images using sliding window approaches. 8. **Motion Analysis and Tracking** ● Principles and methods for analyzing motion and tracking objects in video sequences. ● Human Action Recognition ● Techniques for recognizing and understanding human actions from video data. ● Image Retrieval ● Methods for retrieving images based on content similarity using feature descriptors and indexing techniques.
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group projects, and Term papers presentations and discussions.
Attendance requirements	Regular attendance is required
Textbooks and references	1. ComputerVision: Algorithms and Applications, by Richard Szeliski, Springer, 2010. (Available free online) 2. Computer Vision: A Modern Approach (2nd edition), by D.A. Forsyth and J. Ponce, Prentice Hall, 2011. 3. Learning OpenCV, by Gary Bradski & Adrian Kaehler, O'Reilly Media, 2008. 4. Pattern Classification (2nd Edition), by R.O. Duda, P.E. Hart, and D.G. Stork, Wiley-Interscience, 2000.

Addis Ababa Science and Technology University Artificial Intelligence and Robotics Center of Excellence				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6022	**Deep Learning**	3	2	3
Class year/Semester	Year I/Semester II			
Pre-requisites course(s)	None			
Course Owner				
Course Rationale	**Course Rationale**: The rationale behind this course is to equip students with a robust understanding of deep learning, a critical subfield of artificial intelligence that drives many of today's technological advancements. As deep learning techniques become increasingly integral to solving complex problems in various industries, it is essential for students to grasp both foundational and advanced concepts. This course bridges the gap between theoretical knowledge and practical application, prepARINg students to design, implement, and evaluate deep learning models. By covering a broad spectrum of topics, from basic neural network structures to sophisticated architectures like GANs and VAEs, the course ensures that students are well-versed in the latest developments and methodologies. The emphasis on learning theory, data bias, and meta-learning also fosters critical thinking and the ability to adapt to evolving challenges in the field. Ultimately, this course aims to produce skilled professionals who can contribute to and innovate within the dynamic landscape of artificial intelligence and deep learning.			
Course objectives and competencies to be acquired	**Course objectives:** Upon the successful completion of the course students will be able to: ● Understand the historical and modern definitions of artificial intelligence and the current role of machine and deep learning. ● Acquire a foundational knowledge of probability as it applies to AI and deep learning. ● Differentiate between generative and discriminative methods, and understand the principles of perceptrons, artificial neural networks, support vector machines, and kernel functions.			

	● Develop proficiency in advanced deep learning architectures, including convolutional neural networks (CNNs), stochastic gradient descent (SGD), and batch normalization techniques. ● Explore and implement generative models such as generative adversarial networks (GANs), deterministic autoencoders, variational autoencoders (VAEs), flow-based models, and self-supervised learning. ● Gain expertise in sequential models, including recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and gated recurrent units (GRUs). ● Understand learning theory, recognize and address data bias, and explore concepts of meta-learning (learning to learn).
Course description/content	***Course Description***: "Introduction to Deep Learning" provides a comprehensive exploration of the key concepts and techniques in the field of deep learning. The course begins with an overview of artificial intelligence, tracing its historical definitions and modern roles, emphasizing the significance of machine and deep learning in today's technological landscape. Students will gain fundamental knowledge of probability and delve into generative and discriminative methods, including perceptrons, artificial neural networks, support vector machines, and kernel functions. The course covers active learning and progresses to advanced topics such as convolutional neural networks, stochastic gradient descent, and batch normalization. Students will explore generative adversarial networks, deterministic and variational autoencoders, flow-based models like PixelCNN and Wavenet, and self-supervised learning. Additionally, the course examines recurrent neural networks, including LSTM and GRU architectures. The learning theory component addresses data bias and meta-learning (learning to learn), providing a holistic view of the principles underpinning deep learning. **Course Content** **Artificial Intelligence Foundations** ● Historical and modern definitions of artificial intelligence ● Current role of machine learning and deep learning ● Fundamental concepts of probability **Core Machine Learning Methods** ● Generative and discriminative approaches ● Perceptrons and artificial neural networks ● Support vector machines and kernel functions ● Principles of active learning **Deep Learning Architectures** ● Convolutional neural networks (CNNs) ● Stochastic gradient descent (SGD) ● Techniques for batch normalization **Advanced Generative Models** ● Generative adversarial networks (GANs) ● Deterministic and variational autoencoders (VAEs) ● Flow-based models (e.g., PixelCNN, Wavenet)

	● Self-supervised learning methods **Sequential Models** ● Recurrent neural networks (RNNs) ● Long short-term memory (LSTM) networks ● Gated recurrent units (GRUs) **Learning Theory and Challenges** ● Theoretical foundations of learning ● Addressing data bias in models ● Concepts of meta-learning (learning to learn)
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	● Deep Learning, 1st Edition, Ian Goodfellow, Yoshua Bengio, and Aaron Courville, MIT Press, 2016, ISBN No. 0262035618. Available at: https://deeplearningbook.org ● Josh Patterson and Adam Gibson (2017). Deep learning: a practitioner's approach. O'Reilly. ● Andreas C. Muller and Sarah Guido (2017). Introduction to machine learning with python. O'Reilly. ● Mohamed Elgendy (2020). Deep learning for vision systems. O'Reilly

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Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6023	**Probabilistic Graphical Models**	3	2	3
Class year/Semester	Year II/Semester I			
Pre-requisites course(s)	Students are expected to have background in basic probability theory, statistics, programming, algorithm design and analysis,			
Course Owner				

Course Rationale	**Course Rationale**: The course provides students with a deep understanding of how probability theory and graph theory can be combined to model complex, uncertain systems. By learning how to represent relationships between variables using graphical structures like Bayesian Networks and Markov Networks, students will gain powerful tools for reasoning under uncertainty, making predictions, and learning from data. The course covers key topics such as inference, learning algorithms, and applications in real-world AI problems, equipping students with essential skills for solving complex tasks in fields like robotics, computer vision, and natural language processing.
Course objectives and competencies to be acquired	**Course objectives** ● The aim of this course is to develop the knowledge and skills necessary to design, implement and apply these models to solve real problems. The course will cover: (1) Bayesian networks, undirected graphical models and their temporal extensions; (2) exact and approximate inference methods; (3) estimation of the parameters and the structure of graphical models.
Course description/content	**Course description**: Probabilistic graphical models are a powerful framework for representing complex domains using probability distributions, with numerous applications in machine learning, computer vision, natural language processing and computational biology. Graphical models bring together graph theory and probability theory, and provide a flexible framework for modeling large collections of random variables with complex interactions. This course will provide a comprehensive survey of the topic, introducing the key formalisms and main techniques used to construct them, make predictions, and support decision-making under uncertainty. **Course Content** • Introduction, Probability Theory, Bayesian Networks • Undirected models • Learning Bayes Nets • Exact Inference; Message Passing • Sampling • MAP Inference; Structured prediction • Parameter Learning • Bayesian Learning; Structure Learning • Exponential families; variational inference

Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	Probabilistic Graphical Models: Principles and Techniques by Daphne Koller and Nir Friedman. MIT Press. https://ermongroup.github.io/cs228-notes/ https://cs.stanford.edu/~ermon/cs228/index.html Further Readings: Modeling and Reasoning with Bayesian networks by Adnan Darwiche. Pattern Recognition and Machine Learning by Chris Bishop. Machine Learning: a Probabilistic Perspective by Kevin P. Murphy. Information Theory, Inference, and Learning Algorithms by David J. C. Mackay. Available online. Bayesian Reasoning and Machine Learning by David Barber. Available online. Graphical models, exponential families, and variational inference by Martin J. Wainwright and Michael I. Jordan. Available online

20. Elective Course Description

This section introduces elective course description that delves into specialized topics tailored to the interests and career aspirations of students. Elective courses offer opportunities to explore subjects beyond core curriculum requirements, allowing for deeper engagement with specific fields or disciplines. These courses often provide students with the flexibility to pursue areas of personal interest, expand their knowledge base, and develop specialized skills that complement their academic journey. Elective Course Description sections typically outline the course's objectives, content, prerequisites, and the intended learning outcomes, offering students insights into what they can expect and achieve through their elective studies.

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Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6024	**Cognitive Modeling**	3	2	3

Class year/Semester	Year II/Semester I
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationale**: The course is designed to equip students with a profound understanding of computational and mathematical approaches to studying human cognition and perception. As cognitive sciences advance, there is a pressing need to comprehend the intricate mechanisms of human thought processes and their replication in artificial systems. This course bridges theoretical knowledge with practical applications, prepARINg students for contributions in both scientific research and technological innovation. By exploring a wide range of modeling paradigms—declarative, connectionist, Bayesian, and dynamical systems—students will gain essential tools for interpreting behavioral, neuropsychological, and neuroscientific data and developing techniques to address cognitive deficits and enhance learning. Emphasizing the application of cognitive models to artificial intelligence, the course highlights how human cognition insights can inform sophisticated AI systems' development. Additionally, topics such as the physics of cognition and quantum models encourage interdisciplinary exploration, fostering innovative approaches. By the end of the course,
Course objectives and competencies to be acquired	At the end of the course the learner will be able to: ● Understand the mechanisms of information processing in the human brain. ● Interpret behavioral, neuropsychological, and neuroscientific data using cognitive models. ● Develop techniques for remediation of cognitive deficits due to brain injury and developmental disorders. ● Facilitate learning processes in normal cognition through cognitive modeling. ● Construct computer architectures that mimic human-like intelligence. ● Explore various modeling paradigms in cognitive science:

	declarative/logic-based, connectionist, Bayesian, and dynamical systems. ● Model episodic and semantic memory computationally, and explore psycholinguistics with an emphasis on lexical semantics. ● Investigate classical models of rationality, symbolic reasoning, decision-making, inductive generalization, causality, categorization, similarity, and the role of analogy in problem-solving. ● Study cognitive development including child concept acquisition, language learning, and arithmetic skill acquisition. ● Analyze the impact of modeling human cognition on Artificial Intelligence, exploring cognitive architectures like ACT-R, SOAR, OpenCog, CopyCat, and Memory Networks. ● Examine interdisciplinary perspectives including the physics of cognition, quantum models of cognition, emergence inspired by statistical physics, and non-linear dynamics in cognition.
Course description/content	**Course Description:** Cognitive modeling involves the design of computer simulations and mathematical models of human cognition and perception. The goals of cognitive modeling include understanding mechanisms of information processing in the human brain, interpreting behavioral, neuropsychological, and neuroscientific data, suggesting techniques for remediation of cognitive deficits due to brain injury and developmental disorders, facilitating learning in normal cognition, and constructing computer architectures to mimic human-like intelligence. The range of modeling tools in cognitive science is vast, including production systems, neural networks, Bayesian probabilistic models, and pure mathematical theories. **Course Content:** **Chapter 1: Introduction to Cognitive Modeling** ● The journey from machines that compute to machines that "think" ● Central challenges and philosophical conundrums ● Schools of thought and the big debates **Chapter 2: Modeling Paradigms** ● Declarative/logical-based computational cognitive modeling ● Connectionist models of cognition ● Bayesian models of cognition ● Dynamical systems approach to cognition **Chapter 3: Cognitive Models of Memory and Language** ● Computational models of episodic and semantic memory ● Modeling psycholinguistics with an emphasis on lexical semantics ● Towards deep understanding: modeling the interaction of language, memory, and learning **Chapter 4: Modeling Select Aspects of Cognition** ● Classical models of rationality, symbolic reasoning, and decision-making ● Formal models of inductive generalization, causality, categorization, and similarity

	● The role of analogy in problem-solving **Chapter 5: Cognitive Development** ● Child concept acquisition ● Child language learning ● Acquisition of arithmetic skills **Chapter 6: Cognition and Artificial Intelligence** ● The impact of modeling aspects of human cognition on Artificial Intelligence ● Cognitive architectures such as ACT-R, SOAR, OpenCog, CopyCat, Memory Networks ● Great ideas and open problems: revisiting the gap **Chapter 7: Physics of Cognition** ● Exploring the parallels between the science of matter and the science of information ● Quantum models of cognition ● Models of emergence inspired by statistical physics ● Non-linear dynamics and cognition
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	1. Bainbridge, L., & Sanderson, P. (1995). Verbal protocol analysis. In J. Wilson & C. Corlett (Eds.), Evaluation of human work (2nd ed.). 411-436. London: Taylor & Francis. 2. Carley, K., Park, D., & Prietula, M. (1993). Agent honesty, cooperation and benevolence in an artificial organization. In Proceedings of workshop on AI and theories of groups & organizations. . Washington, DC: 3. Congdon, C. B., & Laird, J. E. (1995). The Soar user's manual, version 7. Ann Arbor, MI: Electrical Engineering and Computer Science Department, U. of Michigan. 4. Grant, D. A. (1962). Testing the null hypothesis and the strategy and tactics of investigating theoretical models. Psychological Review, 69(1), 54-61. 5. Gray, W. D., John, B. E., & Atwood, M. E. (1992). The precis of Project Ernestine or An overview of a validation of GOMS. In CHI'92 Conference on Human Factors and Computing Systems. New York, NY: ACM Press. 6. Gray, W. D., John, B. E., & Atwood, M. E. (1993). Project Ernestine: Validating a GOMS analysis for predicting and explaining real-world task performance. Human- Computer Interaction, 8(3), 237-309.

Addis Ababa Science and Technology University Artificial Intelligence and Robotics Center of Excellence				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6025	**Emerging AI Technologies**	3	2	3

Class year/Semester	Year I/Semester II
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationales:** This course is essential in today's context where the proliferation of AI technologies significantly impacts the environment and society. It aims to sensitize students to the sustainability challenges posed by AI systems and the importance of designing eco-friendly solutions. By focusing on the principles of sustainable AI, the course ensures that students can critically assess and mitigate the negative impacts of AI technologies. Moreover, the course emphasizes the integration of computational and sustainability sciences, fostering a holistic approach to AI development. The goal is to prepare students to be conscientious AI practitioners who prioritize sustainability and ethical considerations in their work, thereby contributing positively to the advancement of technology and society.
Course objectives and competencies to be acquired	**Course objectives** On successful completion of this course the learner will be able to: ● Identify, synthesize, and communicate the impacts of non-green AI designs, models, algorithms, and components on humanity, society, and environment. ● Critically assess AI systems in terms of parameters inherently related to sustainability. ● Design and evaluate sustainable AI strategies to bridge computational and sustainability science for assessing, evaluating, designing, and deploying greener AI models and products. ● Demonstrate expert knowledge on usable, trusted, reproduceable AI development.
Course description/content	**Course Description:** The course delves into the intersection of artificial intelligence and sustainability, aiming to educate students on the environmental and societal impacts of AI technologies. Students will learn to identify, synthesize, and communicate the effects of non-green AI designs, models, algorithms, and components. The course

	provides a framework for critically assessing AI systems through a sustainability lens and equips students with the skills to design and evaluate sustainable AI strategies. Emphasis is placed on developing AI models and products that are usable, trusted, and reproducible, ensuring ethical and responsible AI development. By the end of the course, students will possess expert knowledge in creating and implementing greener AI solutions. **Course Content** **Red and Green AI** ● Measures of Efficiency; Carbon emission; electricity usage; Number of learning parameters; Cost of AI development; FPO; FPO Cost of Existing Models; **Sustainability principles for AI** ● Energy and Policy Considerations for AI or Deep Learning; Recommendations; Need for protocols to record how much energy is used when training an AI model; Guidelines for defining performance gain; Policy and recommendations for Big Data for sustainability and sustainable future, respectively. **Towards Greener AI Technology** ● Better algorithm designs; Better memory management; Ways to reduce computing power in terms of consumption of energy and space. Better hardware design for efficient deep learning algorithms; empirical justification of model complexity; conceptual or practical simplification of an existing model in terms of interpretability, inference time, robustness **Artificial Intelligence for Social Good** ● Encouraging technology affordable by SMEs; AI technology having financial impacts as well as environment and/or societal impacts [e.g., public health, education]; AI (models) for public commitments. **Machine Learning and Decision Making for Sustainability** ● Bridging computational science and sustainability science; advanced learning techniques for this purpose, e.g., transfer learning (TL), knowledge distillation (TL for data gaps; KD for smaller models from larger models); Pruning bigger models; quantization for model compression **Bias Awareness and Mitigation** ● Model/algorithmic bias; Bias from training data, e.g., gender bias; algorithms for debiasing models; Fairness of AI; identifying and preventing discriminatory action of AI; **Refocusing on common sense** ● Emphasizing on societal and environmental impacts of AI; safety (self-driving cars); threats of AI to the future of humanity (autonomous weapons; weapons of destruction); ● Encouraging alternative areas of AI research; empowering marginalized research areas and researchers ● Empower marginalized researchers (e.g., engaging women and

	minority research groups) and areas; **Emerging AI technology and sustainability I** ● AI for sustainability; Environmental sustainability; AI for climate change; AI, smart and sustainable cities; ModelingModelling Commuting Patterns; Vision-Based Road Traffic Congestion Monitoring; Disease Surveillance and Diagnosis (e.g., imaging, genomics, electronic health records, drug discovery); AI for reducing food waste. ● Intelligence Gathering, compensating for a lack of human experts; Crop disease monitoring; Identifying drought and agricultural trends in every locality; Prediction of food insecurity from remote sensing data; Cropland disappearance; Sustainable agriculture, wildlife, and agro-engineering **Forecasting, prevention, and mitigation Malicious Use of AI** ● Malicious Use of AI-Based Deepfake Technology; Malicious chatbots; Forecasting, prevention, and mitigation Malicious Use of AI **Facilitating Usable AI and Trusted AI** ● Facilitating usable AI development; facilitating trusted AI development; Emphasize reproducibility.
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	1. Mitchell, Melanie. *Artificial Intelligence: A Guide for Thinking Humans*. Farrar, Straus and Giroux, 2019. 2. Poole, David L., and Alan K. Mackworth. *Artificial Intelligence: Foundations of Computational Agents*. Cambridge University Press, 2017 3. Müller, Vincent C., and Nick Bostrom (Eds.). *Ethics of Artificial Intelligence and Robotics*. Cambridge University Press, 2018. 4. Joshi, Amit, and Nitin Bhatia (Eds.). *Artificial Intelligence for Sustainable Development*. Springer, 2021.

Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6026	**Speech Processing**	3	2	3

Class year/Semester	Year II/Semester I
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationale**: The "Speech Processing for Human-Computer Interaction" course addresses the increasing importance of speech technologies in modern computing interfaces. As voice interfaces become ubiquitous in devices and applications, understanding the fundamental principles of speech production, perception, recognition, synthesis, and dialogue systems is crucial. This course equips students with the knowledge and skills to design and implement effective speech-based interaction systems, catering to diverse user needs and enhancing user experience across various domains.
Course objectives and competencies to be acquired	Upon successful completion of the course the learner will be able to: **Course objectives** ● Gain a comprehensive understanding of the basic principles of sound and speech production, including acoustic theory and sound propagation in the vocal tract. ● Explore fundamental concepts in speech perception, including auditory models, sound perception models, and Mean Opinion Score (MOS) methods. ● Learn key techniques in digital speech processing, such as time domain methods, pitch period estimation, and Short-Time Fourier Transform (STFT) analysis. ● Understand advanced speech processing methods, including Linear Predictive Coding (LPC), homomorphic speech processing, and various speech waveform coding techniques. ● Obtain an introductory overview of speech synthesis, speech recognition, and dialogue systems, and their applications in human-computer interaction contexts.
Course description/content	**Course Description**: This course introduces students to speech processing with a focus on human-computer interaction, covering speech synthesis, speech recognition, and dialogue systems. It aims to provide foundational knowledge in sound and speech production and perception, along with the basic principles underlying speech recognition, synthesis, and dialogue system technologies. Students will gain an introductory overview of digital speech-processing techniques essential for understanding and developing applications in this field. **1. Introduction to Speech Processing** ● Overview of digital speech processing and its applications in human-

	computer interaction. **2. DSP Fundamentals Review** ● Review of fundamental concepts in Digital Signal Processing (DSP) relevant to speech processing. **3. Acoustic Theory of Speech Production** ● Understanding the principles governing sound production and propagation in speech. **4. Speech Perception** ● Models and methods for speech perception, including auditory and sound perception models. **5. Time Domain Methods in Speech Processing** ● Techniques for analyzing speech signals in the time domain. **6. Short-Time Fourier Transform (STFT)** ● Concepts and methods for spectral analysis of speech using STFT. **7. Speech Representations and Synthesis** ● Methods for synthesizing speech based on STFT analysis and synthesis techniques. **8. Linear Predictive Coding (LPC)** ● Principles of LPC, its frequency domain interpretations, and methods for speech synthesis and vocoding. **9. Speech Waveform Coding** ● Various coding methods including uniform and non-uniform quantization, adaptive and differential quantization. **10. Model-Based Speech Coding** ● Approaches to speech coding based on modeling techniques for efficient data representation.
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	1. MCTEAR, Michael. Spoken dialogue technology: toward the conversational user interface. Edited by T. V. Raman. New York: Springer, 2004. xii, 432. ISBN 1852336722. Info 2. https://web.ece.ucsb.edu/Faculty/Rabiner/ece259/speech%20course.html

Addis Ababa Science and Technology University Artificial Intelligence and Robotics Center of Excellence				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN6027	**Explainable AI**	3	2	3

Class year/Semester	
Pre-requisites course(s)	
Course Owner	
Course Rationale	**Course Rationale**: The rationale is to equip students with the critical skills needed to understand, interpret, and justify the decisions made by complex AI models. As AI systems are increasingly deployed in high-stakes fields such as healthcare, finance, and law, the ability to provide transparency and accountability is essential. This course will cover methodologies and techniques that make AI systems more interpretable and trustworthy, fostering ethical AI development and promoting user confidence. It also addresses the legal and regulatory demands for explainability in AI, prepARINg students to meet industry standards.
Course objectives and competencies to be acquired	**Course objectives** After a successful course, the student shall **Knowledge and understanding** - Show familiarity with concepts within Explainable AI and interpretable machine learning - Demonstrate comprehension of current techniques for generating explanations from black-box machine learning methods - Demonstrate comprehension of current ethical, social and legal challenges related to **Explainable AI Skills and abilities** - Demonstrate the ability to select and assess Explainable AI methods **Judgement and approach** - Demonstrate the ability to review, present and critically assess state-of-the-art papers in relevant areas within Explainable AI
Course description/content	**Course description**: This course gives an introduction to Explainable AI (XAI), providing an overview of relevant concepts such as interpretability, transparency and black-box machine learning methods. The course provides an overview of state-of-the-art methods for generating explanations, and touches upon issues related to decision-support, human interaction with AI/intelligent systems and their

	evaluation. In summary, the Explainable AI course covers the following topics: - Definitions and concepts such as black-box models, transparency, interpretable machine learning and explanations. - Decision-making and decision support, Human-Computer Interaction (HCI) and AI. - Explainable AI. - Methods for Explainable AI. - Applications and examples. - Trust and acceptance. - Evaluation methods and metrics. - Ethical, legal and social issues of explainable AI
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	

Addis Ababa Science and Technology University **Artificial Intelligence and Robotics Center of Excellence**				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN7012	**Reinforcement Learning**	3	2	3
Class year/Semester				
Pre-requisites course(s)	Proficiency in Python, Basic Probability and Statistics, Foundations of Machine Learning			
Course Owner				
Course Rationale	**Course Rationale**: The course aims to equip learners with foundational and advanced concepts in this critical area of machine learning, where agents learn to make decisions by interacting with their environment to maximize cumulative rewards. Through a combination of theory and practical application, the course covers key topics such as Markov decision processes, dynamic programming, Monte Carlo methods, temporal difference learning, and deep reinforcement learning. Students will gain the ability to apply these techniques to solve complex, real-world problems in areas like robotics, game AI, and autonomous systems, fostering a deep understanding of how intelligent agents can learn optimal behaviors through trial and error.			

Course objectives and competencies to be acquired	**Course objectives** By the end of the class students should be able to: • Define the key features of reinforcement learning that distinguishes it from AI and non-interactive machine learning (as assessed by the exam). • Given an application problem (e.g. from computer vision, robotics, etc), decide if it should be formulated as a RL problem; if yes be able to define it formally (in terms of the state space, action space, dynamics and reward model), state what algorithm (from class) is best suited for addressing it and justify your answer (as assessed by the exam). • Implement in code common RL algorithms (as assessed by the assignments). • Describe (list and define) multiple criteria for analyzing RL algorithms and evaluate algorithms on these metrics: e.g. regret, sample complexity, computational complexity, empirical performance, convergence, etc (as assessed by assignments and the exam). • Describe the exploration vs exploitation challenge and compare and contrast at least two approaches for addressing this challenge (in terms of performance, scalability, complexity of implementation, and theoretical guarantees) (as assessed by an assignment and the exam).
Course description/content	**Course description**: To realize the dreams and impact of AI requires autonomous systems that learn to make good decisions. Reinforcement learning is one powerful paradigm for doing so, and it is relevant to an enormous range of tasks, including robotics, game playing, consumer modeling and healthcare. This class will provide a solid introduction to the field of reinforcement learning and students will learn about the core challenges and approaches, including generalization and exploration. **Course contents**: • **Introduction to Reinforcement Learning** • Overview of Machine Learning Paradigms • Core Concepts: Agents, Environments, Actions, States, and Rewards • Real-world Applications of Reinforcement Learning • **Markov Decision Processes (MDPs)** • MDP Framework and Formal Definitions • Policy, Value Function, and Bellman Equations • Solving MDPs using Dynamic Programming • **Monte Carlo Methods**

	• Prediction and Control with Monte Carlo Methods • Exploring-Starts and On-Policy Learning • Off-Policy Learning with Importance Sampling • **Temporal Difference Learning** • TD(0), SARSA, and Q-Learning • n-Step Bootstrapping and Eligibility Traces • Exploration vs. Exploitation Dilemma • **Policy Gradient Methods** • Introduction to Policy-based Methods • REINFORCE Algorithm and Actor-Critic Methods • Advantage Actor-Critic (A2C) and Proximal Policy Optimization (PPO) • **Deep Reinforcement Learning** • Combining Neural Networks with RL: Deep Q-Networks (DQN) • Enhancements to DQN: Double DQN, Dueling DQN, and Prioritized Experience Replay • Advanced Deep RL Algorithms: DDPG, A3C, and SAC • **Multi-Agent Reinforcement Learning** • Introduction to Multi-Agent Systems • Cooperative and Competitive Scenarios • Communication and Coordination Strategies • **Hierarchical and Transfer Learning in RL** • Hierarchical Reinforcement Learning (Options Framework) • Multi-task and Transfer Learning in RL • Meta-Reinforcement Learning • **Reinforcement Learning in Robotics** • Application of RL to Robotics and Autonomous Systems • Continuous Control and Policy Learning • Sim-to-Real Transfer Techniques • **Ethics and Challenges in Reinforcement Learning** • Safety and Ethical Implications of Autonomous Agents • Challenges: Sample Efficiency, Stability, and Scalability • Interpretability and Explainability in RL Models • **Project and Case Studies** • Application of RL to a Real-world Problem • Case Studies from Robotics, Finance, Healthcare, and Game AI
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required

Text books and references	• Reinforcement Learning: An Introduction, Sutton and Barto, 2nd Edition. • Reinforcement Learning: State-of-the-Art, Marco Wiering and Martijn van Otterlo,Eds. • Artificial Intelligence: A Modern Approach, Stuart J. Russell and Peter Norvig. • Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville. • David Silver's course on Reinforcement Learning

Addis Ababa Science and Technology University **Artificial Intelligence and Robotics Center of Excellence**				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN**6028**	**Intelligent Robotics**	3	2	3
Class year/Semester	Year I/Semester I			
Pre-requisites course(s)	None			
Course Owner				
Course Rationale	**Course Rationale**: The course is designed to provide students with a comprehensive understanding of the principles, technologies, and methodologies that underpin the development and application of intelligent robotic systems. As robotics continues to revolutionize various sectors, from manufacturing and healthcare to logistics and entertainment, it is crucial for students to acquire the skills and knowledge necessary to design, build, and program sophisticated robots. This course aims to bridge the gap between theoretical foundations and practical implementation, ensuring that students can apply their learning to real-world scenarios. By covering a wide range of topics, including robot mechanics, kinematics, control systems, sensing, perception, and advanced robotics, the course equips students with a holistic view of the field. Emphasis on hands-on projects and the latest industry tools and techniques prepares students to tackle complex challenges in robotics. Furthermore, the course addresses the ethical, societal, and legislative implications of robotics, fostering a responsible approach to technological advancement. By understanding the broader impact of their work, students will be better prepared to develop robots that are not only functional and efficient but also ethical and socially beneficial. Ultimately, the "Intelligent Robotics" course seeks to cultivate a new generation of robotics experts who are adept			

	at both the technical and ethical dimensions of their field, ready to innovate and lead in the dynamic landscape of intelligent robotics.
Course objectives and competencies to be acquired	**Course Description:** This course delves into the science and design of robots—engineered artifacts that integrate computation and physical embodiment to exhibit intelligent behavior. Students will explore the principles and algorithms that enable robots to interact with and navigate the physical world. Key topics include behavior-based embodied AI, kinematics, inverse kinematics, geometric reasoning, motion planning, mapping, manipulation, dynamics, biologically inspired and biomimetic robotics, distributed robotics and intelligence, and philosophical questions about the nature of intelligence in physical entities.
Course description/content	**Course Content:** **Chapter 1: Fundamentals of Robotics** ● Robot Mechanics: Understanding the mechanical structure and functioning of robots. ● Engineering Mechanics: Principles of mechanics applied to robot design and function. ● Electromechanics: Integration of electrical and mechanical systems in robots. ● Electric Drives: Electrical systems that power robot movement. ● Applied Electronics: Electronic components and circuits used in robotics. **Chapter 2: Kinematics and Dynamics** ● Kinematics, Statics, and Dynamics of Robots: Analysis of motion, forces, and torques in robotic systems. ● Robot End Effectors: Design and functionality of tools and devices at the end of a robotic arm. ● Dynamics: Understanding the forces and motions in robotic systems. **Chapter 3: Control Systems and Programming** ● Design of Robots and Their Control Systems: Creating robots and their control mechanisms. ● Control Systems and Programming of Robots: Developing and implementing control algorithms. ● Programming of Robots: Software development for robot control and functionality. ● Real-Time Systems: Systems that process data and respond in real time. ● Automatic Control Theory: Theoretical foundations of automatic control. **Chapter 4: Sensing and Perception** ● Robotic Sensors: Types, limitations, and processing of sensors used in robotics. ● Image Processing and Recognition: Techniques for processing and interpreting visual data. ● Geometric Reasoning: Spatial understanding and manipulation in robotics.

	● Mapping: Techniques for representing and navigating environments. **Chapter 5: Advanced Robotics Topics** ● Motion Planning: Algorithms for robot movement and navigation. ● Manipulation: Techniques for controlling robot interaction with objects. ● Biologically Inspired and Biomimetic Robotics: Design principles drawn from biological systems. ● Distributed Robotics and Intelligence: Coordination and communication among multiple robots. ● Philosophical Questions: Exploring the nature of intelligence in physical entities. **Chapter 6: Application Development and User Interaction** ● Intelligent Robot Applications Development Tools: Software and hardware tools for developing intelligent robotic applications. ● User-Centered Design in Robotics: Designing robots with user needs and preferences in mind. ● User Engagement and Perception: Understanding how users interact with and perceive robots. ● Technologies and Languages for Robotics Applications: Programming languages and technologies for building robotic applications. **Chapter 7: Projects, Ethics, and Future Trends** ● Robotic System Projects: Hands-on projects designing and building robotic systems. ● Modeling and Experimental Analysis: Creating and testing models of robotic systems. ● Modeling of Robotized Systems: Simulation and analysis of systems involving multiple robots. ● Computational Intelligence Methods: AI techniques applied to robotics. ● Legislative, Ethical, and Social Implications: Legal, ethical, and societal considerations of robotics. ● Human-Robot Interaction Research: Formulating hypotheses and designing experiments to study interactions between humans and robots.
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentations and discussion.
Attendance requirements	Regular attendance is required

Text books and references	1. Maja Mataric. The Robotics Primer. MIT Press 2007. 2. Roland Siegwart & Illah R. Nourbakhsh. Introduction to Autonomous Mobile Robots. MIT Press 2004. 3. Ronald Arkin. Behavior-Based Robotics. MIT Press 1998. 4. Howie Choset et al. Principles of Robot Motion. MIT Press 2005. 5. John Craig. Introduction to Robotics: Mechanics and Control, 3rd ed. Prentice Hall 2004.

Addis Ababa Science and Technology University **Artificial Intelligence and Robotics Center of Excellence**				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN7013	Neurosymbolic AI	3	2	3
Class year/Semester	**Year II - Semester I**			
Pre-requisites course(s)				
Course Owner				
Course Rationale	**Course Rationale**: The course is designed to address the evolving landscape of Artificial Intelligence, where deep learning models have achieved remarkable success in areas like image recognition, natural language processing, and autonomous systems. However, current limitations in deep learning, such as lack of generalizability, explainability, and common-sense reasoning, highlight the need for a more comprehensive approach. This course advocates for the integration of deep learning with symbolic AI, an earlier method based on logic and probabilistic reasoning, to create hybrid systems that overcome these challenges. By exploring both fast, intuitive decision-making (deep learning) and slow, deliberate reasoning (symbolic AI), students will gain a holistic understanding of AI architectures. This integration mirrors human cognitive processes and offers the potential for more adaptable, robust, and explainable AI systems, making the course essential for those seeking to advance AI research and applications.			
Course objectives and competencies to be acquired	**Course objectives** The course aims to provide students with an in-depth understanding of the complementary roles of deep learning and symbolic AI in modern Artificial Intelligence systems. Students will explore the strengths and limitations of both approaches, focusing on how deep learning models, inspired by biological neural networks, excel in pattern recognition and fast, intuitive decision-making, while symbolic AI offers advantages in areas like generalization, abstraction, explainability, and causal reasoning. By examining how these paradigms can be integrated into hybrid AI systems, the course will equip students to design AI models that are adaptable, robust, and capable of leveraging both rapid learning from data and logical			

	reasoning. Students will also gain insights into the cognitive science behind fast and slow thinking, providing theoretical and practical knowledge to apply these concepts in diverse AI applications.
Course description/content	**Course description**: Much of today's successful Artificial Intelligence models are based on deep learning inspired by biological neural networks. They have eclipsed an earlier approach to AI known as knowledge-based systems in which the world is represented in the form of pre-determined symbols with inference based on logic and probabilistic reasoning. However the symbolic approach can better address current limitations of deep learning, e.g., adaptability, generalizability, robustness, explainability, abstraction, common sense, causal reasoning, etc. The use of both approaches in the same system has cognitive support. Such as fast and slow thinking, wherein deep learning plays the role of fast thinking and the symbolic approach plays the role of slow thinking. The seminar course covers cognitive theories of fast and slow thinking, robust artificial intelligence, parallel and sequential use of deep learning and causal reasoning and implementation issues such as attention and co-operating mulitagents. **Course Content** 1. **Overview** 1. Neuro-symbolic AI Overview 2. Research Topics in Slow and Fast Thinking 3. Robust AI 4. Hybrid Architectures for Robust AI 5. Neuro-symbolic Concept Learner 2. Causality 1. Statistics&Causality 2. Causality in AI 3. Counterfactuals in AI 4. Transfer Learning and Causality 5. Statistical Modeling of Cause and Effect 6. Learning Causal Bayesian Neworks 3. Learning Probabilistic Models 1. Structure Learning in Bayesian Networks 2. Markov Network Parameter Learning 3. Markov Network Structure Learning 4. Knowledge Graphs 1. Knowledge Graphs 2. Statistical Relation Learning 3. MRFs from Knowledge Graphs
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and	https://cedar.buffalo.edu/~srihari/CSE701/index.html

references	

Addis Ababa Science and Technology University **Artificial Intelligence and Robotics Center of Excellence**				
Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN7016	**AI Ethics and Society**	3	2	3
Class year/Semester	Year II/Semester I			
Pre-requisites course(s)	None			
Course Owner				
Course Rationale	**Course Rationales:** As AI technologies become increasingly integrated into various aspects of society, understanding their ethical implications is crucial. This course is designed to equip students with the knowledge and skills to navigate the complex ethical landscape of AI. By grounding students in major ethical theories and frameworks, the course ensures they can critically assess AI technologies and address issues such as privacy, bias, and fairness. The emphasis on real-world applications and current regulations provides students with practical insights into the challenges of governing AI. Moreover, by exploring the roles and values of different stakeholders, students will be prepared to engage in informed discussions and make ethically sound decisions. Ultimately, this course aims to develop ethical leaders who can guide the responsible development and deployment of AI technologies in society.			
Course objectives and competencies to be acquired	Upon successful completion of the course the learner will be able to: ● Gain a foundational understanding of major ethical theories and frameworks that are applicable to AI, such as utilitarianism, deontological ethics, virtue ethics, and care ethics. ● Develop the ability to identify and analyze key ethical issues arising from AI technologies, including privacy concerns, bias and fairness, transparency, accountability, and the potential impacts on employment. ● Enhance critical thinking skills and ethical reasoning abilities to evaluate AI technologies from multiple ethical perspectives, and formulate well-argued positions on AI-related issues. ● Understand how AI technologies impact various sectors of society, including healthcare, criminal justice, finance, and education, and the associated ethical implications. ● Acquire knowledge about current policies, regulations, and			

	guidelines concerning AI, both domestically and internationally, and understand the challenges in regulating AI. ● Learn to analyze the roles, interests, and values of different stakeholders involved in the creation, deployment, and governance of AI technologies. ● Apply ethical principles and frameworks to real-world case studies involving AI, proposing solutions and strategies for ethical dilemmas. ● Develop the ability to effectively communicate ethical analyses and recommendations concerning AI to diverse audiences, including technologists, policymakers, and the general public. ● Explore future trends in AI development and anticipate potential new ethical challenges and societal impacts.
Course description/content	**Course Description:** The "AI Ethics and Society" course provides a comprehensive exploration of the ethical dimensions and societal impacts of artificial intelligence. Students will gain a foundational understanding of major ethical theories and frameworks, such as utilitarianism, deontological ethics, virtue ethics, and care ethics. The course emphasizes the identification and analysis of key ethical issues arising from AI technologies, including privacy, bias, fairness, transparency, accountability, and the impacts on employment. Through critical thinking and ethical reasoning, students will evaluate AI technologies from multiple perspectives and formulate well-argued positions on AI-related issues. The course also covers the impact of AI on various sectors, current policies and regulations, the roles of different stakeholders, and the application of ethical principles to real-world case studies. By the end of the course, students will be able to communicate ethical analyses and recommendations effectively and anticipate future ethical challenges in AI development. **Course Content:** **1. Introduction to AI Ethics** ● Overview of Artificial Intelligence ● Introduction to ethics in technology ● Historical perspective on AI development **2. Ethical Theories and Frameworks** ● Major ethical theories (Utilitarianism, Deontological Ethics, Virtue Ethics) ● Ethical decision-making frameworks applied to AI **3. Privacy and Surveillance** ● AI in data collection and surveillance ● Privacy concerns with AI applications ● Case studies: Social credit systems, facial recognition technology **4. Bias and Fairness** ● Sources of bias in AI systems ● Impact of bias on societal fairness ● Strategies for mitigating bias in AI **5. AI and Employment**

	● Automation, job displacement, and the future of work ● Economic impacts of AI on different sectors ● Universal Basic Income and other societal responses **6. AI in Governance and Decision-Making** ● Use of AI in public policy and governance ● Transparency and accountability in AI systems ● Case studies: AI in legal judgments, electoral processes **7. AI and Military Ethics** ● Autonomous weapons systems ● Ethical implications of AI in warfare ● International laws and AI warfare **8. Robot Rights and Personhood** ● Philosophical questions about AI consciousness and sentience ● Rights of robots and AI systems ● Ethical personhood **9. Global Perspectives on AI Ethics** ● Cultural variations in perceptions of AI ethics ● Global governance of AI technology ● International collaborations and treaties **10.Future of AI Ethics** ● Emerging technologies and future ethical dilemmas ● Role of ethicists in AI development ● Building ethical AI systems: Challenges and strategies
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	1. O'neil, Cathy. Weapons of math destruction: How big data increases inequality and threatens democracy. Crown, 2017. 2. Liao, S. Matthew, ed. Ethics of artificial intelligence. Oxford University Press, 2020. 3. Mitchell, Melanie. "Artificial intelligence: A guide for thinking humans." (2019). 4. Schneider, Susan. Artificial you: AI and the future of your mind. Princeton University Press, 2019. 5. Fry, Hannah. Hello world: Being human in the age of algorithms. WW Norton & Company, 2018. 6. Kearns, Michael, and Aaron Roth. The ethical algorithm: The science of socially aware algorithm design. Oxford University Press, 2019.

Course Code	Course Title	Cr. Hr.	Hour	
			Lect.	Tut./Lab
ARIN7017	**Intelligent Search and Games**	3	2	3

Class year/Semester	Year II/Semester I
Pre-requisites course(s)	None
Course Owner	
Course Rationale	**Course Rationale**: The course is designed to equip students with the specialized knowledge and skills necessary to tackle complex challenges in game development and artificial intelligence. As the gaming industry evolves, there is a growing demand for AI systems capable of sophisticated gameplay and innovative game design. This course bridges theoretical insights with practical implementation, prepARINg students to create intelligent agents that can adapt and excel in various gaming environments. By focusing on advanced search techniques, heuristic optimizations, and modern AI methods like MCTS and deep learning, students will gain insights into cutting-edge AI applications in gaming. Moreover, exploring topics such as evolutionary game design, procedural content generation, and puzzle design fosters creativity and innovation in game development practices. Ultimately, this course aims to empower students to become proficient in leveraging AI to enhance game experiences, drive innovation in game design, and meet the evolving demands of the gaming industry.
Course objectives and competencies to be acquired	**Course objectives** ● Upon the successful completion of the o=course the learner will be able to: ● Understand and implement sophisticated search algorithms such as Alpha-beta pruning, Monte Carlo tree search (MCTS), and their variants to develop AI agents capable of strategic decision-making in complex game environments. ● Explore and apply heuristic approaches including history heuristics, Principal Variation Search (PVS), and other optimization techniques to improve AI decision-making processes and efficiency in game playing scenarios. ● Analyze the application of deep learning techniques exemplified by AlphaGo and AlphaZero, exploring their impact on game AI and their potential for advancing autonomous gameplay capabilities. ● Investigate evolutionary game design principles, procedural content generation (PCG), and puzzle design methodologies to foster creativity and innovation in game development, leveraging AI for

	dynamic and engaging player experiences. ● Develop skills in assessing game quality metrics and evaluating AI performance through self-play scenarios, enabling students to enhance AI agents' capabilities and contribute to the evolution of game AI standards.
Course description/content	**Course Description**: This course delves into advanced techniques and methodologies in Game AI, focusing on both theoretical foundations and practical applications. Students will explore a range of topics essential for developing intelligent agents capable of playing and designing games autonomously. Key areas include sophisticated search algorithms like Alpha-beta pruning and Monte Carlo tree search (MCTS), heuristic approaches for decision-making, such as history heuristics and Principal Variation Search (PVS), and the application of deep learning techniques exemplified by AlphaGo and AlphaZero. Additionally, the course covers evolutionary game design, procedural content generation (PCG), and the metrics used to assess game quality, providing a comprehensive understanding of AI's role in game development and design. **Course Content** **Chapter 1: Search Techniques in Game AI** ● Basic search techniques: Alpha-beta pruning, A* algorithm. ● Advanced search techniques: Iterative Deepening A* (IDA*), B* algorithm, transposition tables, retrograde analysis and endgame databases, proof-number search and variants. ● Multi-player search methods: Strategies for games involving multiple players. **Chapter 2: Heuristics and Search Enhancements** ● Heuristics: World representations for games, killer moves heuristic, history heuristic, Principal Variation Search (PVS), windowing techniques, null-moves heuristic, forward-pruning techniques, selective search, Goal-Oriented Action Planning (GOAP). **Chapter 3: Monte Carlo Methods and Game AI** ● Monte Carlo methods: Introduction to Monte Carlo tree search (MCTS), enhancements such as UCT (Upper Confidence Bounds for Trees), and applications in game playing. ● AlphaGo and AlphaZero approaches: Deep learning-based methods for game AI, their evolution and impact. **Chapter 4: Game Design and AI Applications** ● Evolutionary game design: Using evolutionary algorithms for game design and optimization. ● Game quality metrics: Techniques for evaluating the quality and balance of games. ● Self-play evaluation: Methods for evaluating AI agents through self-play scenarios.

	● Procedural content generation (PCG): Algorithms and techniques for generating game content dynamically. ● Puzzle design: Principles and methods for designing engaging and challenging puzzles in games.
Assessment/Evaluation modalities	Reading Assignments, Reviews, Exams, Group project and Term paper, presentation and discussion.
Attendance requirements	Regular attendance is required
Text books and references	1. AI for Games, Third Edition (earlier editions are fine too), by Ian Millington, publisher: CRC Press, ISBN: 978- 1138483972. 2. Presenting to Win, The Art of Telling Your Story, Updated and Expanded Edition (earlier editions are fine too), by Jerry Weissman, publisher: FT Press (Pearson Education), ISBN: 9780137144174. 3. Millington, I. and Funge, J. (2009). Artificial Intelligence for Games, 2nd Edition Morgan Kaufmann Publishers, ISBN: 978-0123747310 4. Russell, S.J. and Norvig, P. (2010). Artificial Intelligence: A Modern Approach, 3rd edition. Pearson Education, New Jersey. ISBN 0-13-207148-7. 5. Yannakakis, G.N. and Togelius, J. (2018) Artificial Intelligence and Games, Springer, Berlin. ISBN 978-3- 319-63519-4 (eBook) 978-3-319-63518-7 (hardcover)

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2. Digital Education Strategy and implementation plan for Ethiopia 2023-2028)
3. Ministry of Science and Higher Education (MoSHE). Home-Grown Collaborative PhD Programme (HCPP), Addis Ababa, 2020.
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Benchmarks for MSc AI Curriculum Development

In developing the MSc Artificial Intelligence (AI) curriculum, benchmarks were analyzed from three institutions:

1. **Adama Science and Technology University (ASTU)**
    - Focuses on integrating AI with practical engineering disciplines.
    - Emphasizes research and practical application
    - Curriculum aligns with the Ethiopian context, addressing local industry needs.
2. **Addis Ababa University (AAU)**
    - Offers advanced AI topics such as Natural Language Processing, Computer Vision, and AI Ethics.
    - Balances academic research with practical skills, preparing students for both academia and industry.
    - Highlights AI's relevance to national priorities
3. **National College of Ireland (NCI)**
    - The MSc in AI focuses on core AI technologies and their applications in business.
    - Includes modules on Machine Learning, Deep Learning, and Data Analytics, along with practical projects.
    - Strong emphasis on employability with career-focused learning and industry collaborations.
    - Link to NCI Course.

The benchmarking process ensured that the developed curriculum incorporates global trends while addressing Ethiopia's specific educational and industrial needs.

Annex 1 - Questionnaire

Guiding Questions for MSc in Artificial Intelligence (AI) Programs at AASTU

Dear Sir/Madam,

AASTU is in the process of designing and launching a new MSc in Artificial Intelligence (AI) program in partnership with EAII. We are interested in obtaining your opinion and suggestions on the courses and research topics to be covered as well as your recommendation about educational resources and lab facilities necessary for the proposed MSc program. Being one of the key stakeholders and main partner of the proposed program, your feedback and recommendations are essential in order to educate and produce highly competent and well-qualified professionals who can accelerate the digital transformation of our industries through the applications of AI and smart solutions. Hence, we are looking for your valuable opinion and suggestions on what should be considered and incorporated into the curriculum of this nationally and globally important program not only in terms of knowledge, skills and attitudes but with respect to research and educational resources required.

To be more specific:

a) We are interested in assessing the existing demand for educating and producing well-qualified professionals in the field of in ML, AI and Robotics;
b) We are also interested in identifying the major competencies and practical skills of AI that are more important for the digital transformation of our industries.

The proposed new MSc programs is planned to be launched in the coming academic year and will be delivered in face-to-face learning methods as well as through different blended learning approaches. We need your opinions on how we can engage and collaborate with you and key stakeholders in launching and running the proposed MSc programs.

Confidentiality and data management

All feedback and information obtained from you will only be used for the purposed curriculum design and development. Any comment or information you have shared with us will be confidential and fully anonymized.

Thank you all for participating in in this important discussion and interview!

Part-A: General and Demographic Information

1. Name of Informant: _____Gender: _____
2. Affiliation (Institution and Department): _____
3. Current Position: _____
4. What is the highest degree you currently hold?
   - □ Diploma/Associate's degree
   - □ Bachelor's degree
   - □ Master's degree
   - □ Doctoral degree
5. How long have you worked in your current position?
   - □ Less than 2 years
   - □ 2 - 5 years
   - □ 6 - 10 years

□ 11 - 20 years

□ More than 20 years

Curriculum Related Questions

1. What do you think about the major challenges and issues that are facing the digital transformation of our industries and business institutions in identifying well-qualified professionals in the area of MSc in AI and Robotics?
2. How modern industries and business institutions can cope up with the critical shortage experts in the field of AI and Robotics in Ethiopia?
3. What are the professional qualification levels that will be more important and useful for effective applications of AI in Ethiopia? (Probe: BSc, MSc or PhD level)?
4. Does your university or institution have any academic or training program related to AI? If yes, at what level of education (BSc, MSc or PhD) and how many students are attending the programs?
5. Do you have some students, employees or colleagues who are interested to join our proposed MSc programs?
    a. If your answer to Q5 is "Yes", would specify their number and the MSc programs in which they might be interested in?
    b. If your answer to Q5 is "No", would you briefly describe why you may not be interested in?
6. What do you think about the role of academic institutions like AASTU in training and providing more qualified professionals in the field of AI?
7. What are important computational skills and digital competencies that the recent University graduates are lacking in relation to AI?
8. What courses, digital competencies and skill sets related to AI would you like to recommend or suggest to be incorporated in our planned Master's program? Please list the top 3 or 5 topics or courses you would like to recommend or suggest.
9. From your institution/industry perspectives, what specific skills and competencies related to AI would be essentially required from the prospective graduates and/or young engineers?

    Probe:

    a) What types of knowledge and technical expertise?

    b) What types of practical skills (e.g. communication, critical thinking)?

c) What types of behaviors or attitudes (e.g. teamwork, socio-cultural awareness)?

d) What would be the exit learning profile from the Masters in AI should look like?

e) What are the major placement opportunities/job positions that will be available for the Masters in AI graduates (you may outline the existing and/or future jobs that might be available in the near future)?

10. Do you have some experiences that you would like to share with us about the challenges of launching and running new MSc programs related to AI?
11. What are the priority research areas and innovations that are most relevant for MSc Program related to AI? Would you outline the top 5 priority research areas or innovations?
12. What kind of smart solutions would you expect from qualified AI engineers and researchers in order to address the practical problems of our industries or business services?
13. Is there anything else that you think is relevant that we haven't asked about? Do you have any additional ideas or recommendations that you would like to suggest to improve the focus and content of the proposed MSc program?

Thank you for your valuable time and important discussions.

Annex-2 Responses

Gender	Age (in years)	Profession	Work experience (in years)	Education level
Male	26 - 30	Robotics Engineer	7 - 9	MSc/MA
Female	26 - 30	Robotics Engineers	1 - 3	BSc/BA
Female	20 - 25	Robotics Engineer	1 - 3	BSc/BA
Male	31 - 35	Researcher	10 - 12	MSc/MA
Female	20 - 25	Computer Science and Engineering	1 - 3	BSc/BA
Male	26 - 30	Electrical engineer	4 - 6	BSc/BA
Male	26 - 30	Robotics Engineer	4 - 6	MSc/MA
Male	26 - 30	Software developer	1 - 3	MSc/MA
Male	20 - 25	AI Geospatial Researcher	1 - 3	BSc/BA
Male	26 - 30	Software Developer	4 - 6	BSc/BA

Area of specialization	Name your institution/organization	Your position in the institution	If you would like to get updates on the curriculum design please leave your email (optional)
Power System and Electronics Engineer	Ethiopian Artificial Intelegince	Expert	
Electronics and Communication Engineering	Ethiopian ARtificial Intelligence Institute	Team Leader	

Electrical and Computer Enginering	Wolaita Sodo University	Expert	dagmawiteshetu9@gmail.com
Sustainable development and Innovation	Ethiopian Artificial Intelligence Institute	Team Leader, Research and Strategy	
AI Geospatial Researcher	Ethiopian Artificial Intelligence Institute	AI Geospatial Researcher	
power and control	Ethiopian Artificial Intelligence	expert	no
Project Management	Ethiopian Artificial Intelligence Institute	Expert	abdiermi13@gmail.com
Software Engineering	Ethiopian Artificial Intelligence	Software Engineer	
Computer Science and Engineering	Adama Science and Technology University	AI Geospatial Researcher	yadasat437@gmail.com
GIS,Web,NLP	AII	Geo-AI developer	

What do you think about the major challenges and issues that are facing the digital transformation of our industries and business institutions in identifying well qualified professionals in the area of MSc in AI and Robotics?	How modern industries and business institutions can cope up with the critical shortage experts in the field of AI and Robotics in Ethiopia?	What are the professional qualification levels that will be more important and useful for effective applications of AI in Ethiopia? (Probe: BSc, MSc or PhD level)?	Does your university or institution have any academic or training program related to AI? If yes, at what level of education (BSc, MSc or PhD) and how many students are attending the programs?
Talent shortage, skill gap,evolving technology	colaboration, investemnt,support for reaserch and institution	Bsc	MSC
Finding qualified professionals in AI and robotics is challenging due to rapidly evolving technology, skill gaps, and educational disparities. The need for a diverse skill set and practical experience, combined with high demand and global competition, complicates recruitment efforts. Adapting education and fostering industry partnerships are crucial to addressing these issues.	To tackle the shortage of AI and robotics experts in Ethiopia, industries and institutions should improve education through partnerships with schools, offer local training programs, and support research and innovation. Attracting international talent and aligning academic programs with industry needs will also help build a stronger talent pool.	MSc and PhD	yes, MSc and PhD
Skill Gap	Strengthen Educational Programs	MSc	No
lack of or poor recruitment and talent acquisition system	providing trainig short term training for immediate need to providing formal education like BSC, MSC AND PhD	Short term training, BSc, MSc or PhD level	we provide a variety of training programs ranging from short term summercamp training for elementary and

			highschool students to startup firms and PhD program in partnership with AASTU
The major challenges in digital transformation for industries and business institutions, particularly in identifying well-qualified professionals for MSc programs in AI and Robotics, include the rapid pace of technological change, a shortage of skilled talent, the evolving nature of AI and robotics requiring interdisciplinary expertise, and the gap between academic training and industry needs. Additionally, there is a challenge in ensuring diversity and inclusion in these fields, as well as bridging the gap between theoretical knowledge and practical, industry-relevant skills.	Modern industries and business institutions in Ethiopia can cope with the critical shortage of AI and Robotics experts by investing in local talent development through partnerships with universities, offering specialized training programs, and promoting STEM education at earlier stages. They can also collaborate with international experts, leverage remote work opportunities to access global talent, and create incentives to retain skilled professionals within the country.	For effective AI applications in Ethiopia, MSc and PhD levels are particularly important, as they provide advanced expertise in AI, critical thinking, and research skills necessary for innovation. However, BSc levels are also valuable for foundational knowledge and supporting roles in AI implementation.	Yes, my institution offers academic and training programs related to AI at the BSc, MSc, and PhD levels. The number of students attending these programs varies by level, with hundreds enrolled across all levels, reflecting strong interest and demand in the field.
Skills & Interdisciplinary Knowledge Gap, Quality of Education,Industry-Academia Mismatch: a disconnect between what is taught in academic programs and the skills required by industry.	Investment in Education and Training, Partnerships and Collaboration, good Government Policy and Support and Public Awareness and Early Education.	BSc Level Role Focus: Entry-level positions such as AI software developers, data analysts, machine learning engineers, and AI support technicians. MSc Level Role Focus: Mid-level positions such as AI specialists, data scientists, machine learning engineers, and AI project managers. PhD Level Role Focus: High-level positions such as AI researchers, academic professors, AI strategists, and R&D leaders.	yes at phD for 5 tudents and for elementary and high school students.
Identifying well-qualified AI and Robotics professionals is challenging due to rapidly evolving technology, diverse curricula, skills gaps, talent shortages, and assessment difficulties. To address these issues, collaboration between organizations and educational institutions is	To address the AI and Robotics expert shortage in Ethiopia, industries and institutions should invest in local talent development, attract international experts, leverage online resources, foster public-private partnerships, and prioritize continuous learning.	While a BSc degree provides a solid foundation in AI, an MSc or PhD level qualification is often more valuable for effective applications in Ethiopia. These higher degrees offer specialized knowledge, research skills, and the ability to tackle complex AI problems.	Yes, our institutions Ethiopian Artificial Intelligence Institute have PHD program with collaboration with Adama science and technology university.

crucial, along with innovative assessment methods and a focus on practical training.			
The digital transformation of industries and business institutions in Ethiopia faces significant challenges, especially in identifying well-qualified professionals with an MSc in AI and Robotics. These challenges can be broadly categorized into educational, economic, infrastructural, and cultural issues: 1. Educational Challenges Limited Availability of Specialized Programs: There are relatively few higher education institutions offering specialized MSc programs in AI and Robotics. This limits the number of graduates entering the workforce with the necessary expertise. Outdated Curriculum: Many educational programs may not keep pace with the rapidly evolving field of AI and Robotics, resulting in graduates who are not fully equipped with the latest skills and knowledge. Lack of Research Opportunities: There may be limited opportunities for students to engage in cutting-edge research, which is crucial for developing deep expertise in AI and Robotics. Quality of Education: The quality of education in AI and Robotics can vary significantly, with some programs lacking the resources, faculty expertise, and industry partnerships needed to provide a world-class education. 2. Economic Challenges Brain Drain: Talented professionals in AI and Robotics may seek better	Addressing the shortage of AI and robotics experts in Ethiopia requires a multifaceted approach, involving government, educational institutions, private sectors, and international partnerships. Here are some strategies that modern industries and business institutions in Ethiopia can adopt: 1. Investment in Education and Training University Programs: Strengthening AI and robotics curricula in universities and technical institutes is crucial. Collaborations with international universities can help develop advanced courses and research programs. Short-term Training: Industries can sponsor or conduct short-term certification courses in AI and robotics, targeting recent graduates or professionals from related fields. Online Learning: Promoting online courses and partnerships with platforms like Coursera, edX, or Udacity can help workers upskill in AI and robotics. 2. Public-Private Partnerships Collaborative R&D: Businesses can collaborate with universities and research institutions to create innovation hubs or labs focused on AI and robotics. Internships and Apprenticeships: Creating internship programs that give students and young professionals hands-on experience with AI and robotics projects can help bridge the skill gap. 3. Attracting International	MSc or PhD level)	yes, By PhD level, 5 students are attending

opportunities abroad, where they can find higher salaries, more advanced research facilities, and greater career prospects. High Cost of Education: The cost of obtaining an MSc in AI and Robotics, especially abroad, can be prohibitively expensive for many students, limiting access to advanced education in these fields. Lack of Industry Investment: Insufficient investment by local industries in AI and Robotics research and development (R&D) can hinder the creation of job opportunities that attract and retain qualified professionals. 3. Infrastructural Challenges Limited Access to Technology: The availability of advanced technological tools and infrastructure, such as high-performance computing, robotics labs, and AI development environments, may be limited, hindering both education and practical application. Poor Internet Connectivity: Reliable and high-speed internet access is essential for modern AI and Robotics work, including access to global knowledge resources and collaboration with international peers. Poor connectivity can be a significant barrier. Insufficient Data Resources: The development of AI often relies on large datasets, which may not be readily available in Ethiopia. The lack of data infrastructure and data-sharing mechanisms can slow down progress in AI research and application.	Talent Incentives for Experts: Offering competitive salaries, benefits, and conducive working environments to attract international experts can help address immediate shortages. Diaspora Engagement: Engaging Ethiopian professionals in the diaspora who have expertise in AI and robotics can lead to knowledge transfer and collaborative projects. 4. Government Policies and Support Tax Incentives: The government can provide tax incentives to companies investing in AI and robotics education and research. Grants and Funding: Providing grants and funding for AI and robotics startups can stimulate innovation and attract talent. Regulatory Support: Developing clear policies around AI and robotics that encourage innovation while ensuring ethical practices can create a stable environment for growth.		
The real skills, what we are doing in our school is that	By offering better opportunities for those in	PhD(it not about profession but skill)	Yes, I do not know.

some one who had no experience in industry teaches how to work in industry. In addition to this there is no infrastructure and the even if some one is qualified, because there is no enough wage in this area no one is willing to focus on it.	this area both financially and giving enough training while doing real projects.		
well qualified professionals needs to be paid highly. even if they become well qualified professionals inside thier country and they love their country but payment will change thier mind	by supporting fiesible project researches(invest on them)	all level BSc, MSc or PhD level	MSC but they are few

Do you have some students, employees or colleagues who are interested to join our proposed MSc programs? a. If your answer to this question is "Yes", would specify their number and the MSc programs in which they might be interested in? b. If your answer to Q5 is "No", would you briefly describe why you may not be interested in?	What do you think about the role of academic institutions like AASTU in training and providing more qualified professionals in the field of AI?	What are important computational skills and digital competencies that the recent University graduates are lacking in relation to AI?	What courses, digital competencies and skill sets related to AI would you like to recommend or suggest to be incorporated in our planned Master's program? Please list the top 3 or 5 topics or courses you would like to recommend or suggest.
YES, more than 20 student	IT IS RELEVANT	programming profiicency	machine Learning and Deep Learning
yes, In AI program	Academic institutions like AASTU play a crucial role in training and producing qualified professionals in AI. By developing robust AI curricula, offering hands-on training, and fostering research opportunities.	Programming knowladge, ML	ML and DL, Data Science, Robotics and Automation System, AI Policy and Advanced Programing for AI.
Yes, interested MSc in AI and Robotics	Research and Development and Ethical and Responsible AI Training.	Lack of knowledge related to AI knowledge, Small number of lecturer learn AI.	Machine Learning, Robotics, Computer Vision.
5 PhD and more than 200 students and 50+ tech based startups	it can play a key and leading role	they lack practical and market driven knowledge	data science, Machine Learning, IoT and Robotics, computer vision,
a. Yes, I know a few	Academic institutions like	Recent university	Machine Learning &

colleagues who might be Interested in joining your proposed MSc programs. Approximately 3-5 colleagues could be interested, particularly in MSc programs related to AI, Robotics, and Data Science. b. No, I may not be interested due to the potential misalignment with my current career goals and focus areas, or because I am already pursuing advanced studies in a different specialization.	AASTU play a crucial role in training and providing qualified professionals in AI by offering specialized programs, fostering research and innovation, and creating industry partnerships. They equip students with the necessary theoretical knowledge and practical skills, helping to bridge the gap between academia and industry needs. Additionally, these institutions contribute to advancing the field through cutting-edge research and by promoting a culture of continuous learning and development.	graduates often lack practical experience in AI-related computational skills, such as proficiency in programming languages like Python, experience with AI frameworks (e.g., TensorFlow, PyTorch), and hands-on skills in data manipulation, machine learning model development, and deployment. They may also fall short in critical thinking for problem-solving, understanding of ethical AI practices, and the ability to apply AI to real-world industry challenges.	Deep Learning - Covering algorithms, model training, and neural networks. Data Science & Analytics - Focusing on data preprocessing, visualization, and statistical analysis. Robotics & Automation - Addressing robotics systems, control mechanisms, and integration. AI Ethics & Policy - Exploring ethical considerations, regulations, and societal impacts of AI. Natural Language Processing - Including language models, text analysis, and sentiment analysis.
a. i don't konw the exact number but many colleague want to learn specifically machine learning, NLP, data science and analytics and computer vision.	For Research and Innovation, Industry Collaboration, Community and Public Engagement, Support for Startups and Entrepreneurship and Policy Advocacy and National Contribution.	key gaps are the following Programming Proficiency, Data Handling and Management, Machine Learning and Deep Learning Techniques, Software Development Practices, Ethics and Societal Impact of AI.	1) Advanced Machine Learning and Deep Learning 2) Big Data Analytics and Data Engineering for AI 3)Natural Language Processing (NLP) 4) Data Science and Analytics 5) Computer vision
Yes, 14 in Machine learning and robotics	AASTU's role in AI education is crucial for developing a strong and competitive AI workforce in Ethiopia. By offering advanced programs, research opportunities, industry collaborations, and a talent pipeline, AASTU can equip students with the necessary skills and knowledge to contribute significantly to the growth of the AI industry.	Recent university graduates often lack essential computational skills and digital competencies needed for AI applications, including proficiency in programming languages like Python and R, data analysis techniques, machine learning algorithms, and cloud computing platforms.	Machine Learning: Core concepts, algorithms, and applications. Deep Learning: Neural networks, architectures, and practical implementations. Natural Language Processing: Text analysis, language modeling, and applications. Computer Vision: Image and video processing, object recognition, and applications. Reinforcement Learning: Decision-making, agents, and applications.
Yes,	producing high quality professional	machine learning, mathematics behind it and etc	For a Master's program in AI in Ethiopia, it's essential to incorporate courses and skill sets that not only provide theoretical foundations

			but also focus on practical applications relevant to local industries and global trends. Here are the top 5 recommended courses and skill sets: 1. Machine Learning and Deep Learning Course Content: Supervised and unsupervised learning Neural networks and deep learning architectures (CNNs, RNNs, LSTMs, Transformers) Reinforcement learning Model evaluation and tuning Digital Competencies: Hands-on experience with machine learning frameworks like TensorFlow, PyTorch, and Scikit-learn. Competence in building, training, and deploying machine learning models. Importance: This course is foundational for AI, covering the core algorithms and techniques that are widely applied across industries. 2. Data Science and Big Data Analytics Course Content: Data preprocessing and cleaning Exploratory data analysis and visualization Big data technologies (Hadoop, Spark) Statistical methods and probabilistic models Digital Competencies: Proficiency in data manipulation using tools like Python (Pandas, NumPy), R, and SQL. Skills in handling and analyzing large datasets, using cloud platforms (AWS, Google Cloud). Importance: Effective AI relies on large amounts of data. This course equips students with the skills

			needed to manage, process, and analyze data at scale.
Yes, 15	It is best way to change the current problem we are facing now.	Algorithms, programming languages specific to AI and real world projects hand on experience with real and enough training data.	Gen AI, Machine Learning and AI algorithms
yes, Datascience	since univeristies are built for that the role is high	if students interst would be high lacking nothing will not affect since this is 21century	big data, ML and DL and Distributed Systems

From your institution/industry perspectives, what specific skills and competencies related to AI would be essentially required from the prospective graduates and/or young engineers? Probe: a) What types o f knowledge and technical expertise? b) What types of practical skills (e.g. communication, critical thinking)? c) What types of behaviors or attitudes (e.g. teamwork, socio-cultural awareness)? d) What would be the exit learning profile from the Masters in AI should look like? e) What are the major placement opportunities/job positions that will be available for the Masters in AI graduates (you may outline the existing and/or future jobs that might be available in the near future)?	Do you have some experiences that you would like to share with us about the challenges of launching and running new MSc programs related to AI?	What are the priority research areas and innovations that are most relevant for MSc Program related to AI? Would you outline the top 5 priority research areas or innovations?	What kind of smart solutions would you expect from qualified AI engineers and researchers in order to address the practical problems of our industries or business services?	Is there anything else that you think is relevant that we haven't asked about? Do you have any additional ideas or recommendations that you would like to suggest to improve the focus and content of the proposed MSc program?
a	no	Ethical AI and fairness, explianable AI, AI in health center,	ethical and AI farms,explianble AI, AI in health center,autonmous system and robotics,NLP	NO

Knowledge and Technical Expertise, Practical Skills, Behaviors or Attitudes, Exit Learning Profile and Placement Opportunities/Job Position	No	NLP, Computer Vision Robotics and Automation System, AI Policy and Advanced Programing for AI.	Predictive Maintenance:,Automated Customer Support	No
Core AI and Machine Learning Concepts and Robotics and Automation.	i have work in ethiopian artificial intelligence institiute so the thing that may be challenge in my opinion is Faculty Recruitment and Industry Collaboration.	Robotics, Computer Vision, Machine learning, Data Science and Analytics and Natural Language Processing (NLP).	Automation of Routine Tasks and Smart Manufacturing and Maintenance.	Interdisciplinary Approach and Soft Skills Development.
knowledge on AI is critical and it should be supported with soft skills and business skills. since AI is emerging technology, it has a high job potential for future graduates	lack of practical lab and data for study, absence of professors in the area	N/A	they will be well qualified who address community/industry problem and also contribute to the digital transformation of the country	N/A
a) Knowledge and Technical Expertise: Proficiency in machine learning algorithms, deep learning, and data analysis. Expertise in programming languages such as Python and C++. Understanding of AI frameworks and tools (e.g., TensorFlow, PyTorch). Knowledge of robotics, computer vision, and natural language processing. b) Practical Skills: Strong problem-solving and critical thinking abilities. Effective communication for explaining complex technical concepts. Ability to work collaboratively in cross-functional teams. c) Behaviors or Attitudes: Teamwork and collaboration skills.	No, I don't have any experiences.	Explainable AI (XAI): Developing methods to make AI decision-making transparent and understandable to users. Machine Learning Efficiency: Enhancing algorithms for better performance with reduced computational resources. Ethical AI: Addressing issues of fairness, bias, and ethical considerations in AI systems. AI for Healthcare: Innovations in using AI for diagnostics, personalized treatment, and healthcare management. Autonomous Systems: Advancing technologies in	Predictive Analytics: Utilizing machine learning to forecast trends and improve decision-making. Automation: Implementing AI-driven automation to enhance efficiency and reduce operational costs. Personalization: Creating customized customer experiences through AI algorithms. Data Insights: Analyzing big data for actionable insights and strategic planning. Advanced Robotics: Developing intelligent robotic systems for complex tasks and operational support. These solutions can address practical problems by enhancing productivity, optimizing processes, and driving innovation.	No

Adaptability and a willingness to continuously learn. Socio-cultural awareness and ethical considerations in AI applications. d) Exit Learning Profile: Comprehensive understanding of AI and robotics principles. Ability to design and implement AI solutions and robotics systems. Capacity to tackle real-world problems with practical and innovative approaches. e) Placement Opportunities/Job Positions: AI Research Scientist Machine Learning Engineer Robotics Engineer Data Scientist AI Solutions Architect Positions in emerging fields like autonomous systems and AI ethics.		robotics and autonomous vehicles for improved safety and functionality.		
a) programming (machine learning, data science, Robotics engineer,NLP, computer vision). b) communication . problem solving and critical thinking skill, c) team work , positive mind set, hard working d) oundational Knowledge and Understanding, Technical and Computational Skills, Research and Innovation, Industry and Application Knowledge. e) Machine Learning Engineer, data scientist , big data engineer, AI product manager, computer vision engineer, NLP engineer etc.)	Faculty Recruitment and Retention, Resource Allocation, Industry Collaboration and Internships, Accreditation and Quality Assurance and Adapting to Technological Advancements.	1) Explainable AI (XAI) 2)Autonomous Systems and Robotics 3) Natural Language Processing (NLP) and Understanding 4) AI for Healthcare and Biomedical Applications 5) AI Ethics and Fairness	1) Predictive Analytics for Business Optimization 2) Automated Customer Support and Service 3) Smart Manufacturing and Predictive Maintenance 4) Healthcare Diagnostics and Personalized Medicine 5) Smart Agriculture and Precision Farming 6) Energy Management and Optimization.	nothing I have to add thanks for inquiry!
AI graduates should possess a strong	Challenges in launching and	Generative AI, Explainable AI, AI	AI engineers and researchers can offer smart	No, I haven't

foundation in machine learning, deep learning, natural language processing, and computer vision. Technical expertise should include proficiency in programming languages like Python, data analysis tools, and cloud platforms. Practical skills such as problem-solving, critical thinking, communication, and teamwork are essential. Graduates should also demonstrate a growth mindset, adaptability, and a passion for innovation. The exit learning profile should reflect these skills and competencies, enabling graduates to contribute effectively to AI research, development, and applications. Potential job opportunities include AI researchers, data scientists, machine learning engineers, software developers, and AI consultants.	running new AI MSc programs include curriculum development, faculty recruitment, resource allocation, student preparedness, industry partnerships, and program evaluation.	for Social Good, Ethical AI and Accelerating AI computations with specialized hardware and cloud platforms.	solutions like predictive analytics, process automation, personalized recommendations, natural language processing, computer vision, anomaly detection, optimization, and decision support to address practical problems in industries and business services.	
a) Types of Knowledge and Technical Expertise For prospective graduates and young engineers in AI, the following types of knowledge and technical expertise are essential: Core AI and Machine Learning Algorithms: Understanding of supervised, unsupervised, and reinforcement learning. Expertise in neural networks, deep learning architectures (CNNs, RNNs, Transformers). Knowledge of	No	For an MSc program related to AI, it's important to focus on research areas and innovations that address both global trends and local challenges. Here are the top 5 priority research areas and innovations relevant for such a program: 1. AI for Agriculture and Food Security Focus: Developing AI solutions for precision agriculture, crop monitoring, pest detection, and yield prediction.	Qualified AI engineers and researchers can develop a variety of smart solutions to address practical problems in industries and business services in Ethiopia. These solutions should be tailored to the local context, taking into consideration the specific challenges and opportunities within the country. Here are some key types of smart solutions that could be expected: 1. Predictive Maintenance Systems Application: Manufacturing, Energy, Transportation Solution: AI-driven predictive	No

probabilistic models, decision trees, and ensemble methods. Data Science and Statistical Analysis: Proficiency in data manipulation, cleaning, and preprocessing techniques. Strong grasp of statistical methods and probability theory. Knowledge of data visualization tools and techniques. Programming and Software Development: Mastery of programming languages such as Python, R, C++, and Java. Experience with AI frameworks and libraries like TensorFlow, PyTorch, and Scikit-learn. Understanding of software engineering principles, including version control, testing, and deployment. Domain-Specific Knowledge: Application of AI in specific industries such as healthcare, finance, agriculture, manufacturing, or energy. Understanding of how AI can address local challenges in Ethiopia, such as language processing for local languages, agricultural automation, or healthcare diagnostics. AI Ethics and Policy: Knowledge of ethical considerations in AI development and deployment. Familiarity with global and local regulations regarding data privacy, security, and AI		Relevance: Agriculture is a critical sector in Ethiopia, and AI can play a transformative role in improving productivity and food security. Research could explore the use of machine learning models to predict crop yields, optimize irrigation systems, or develop AI-driven tools for early pest and disease detection. Innovation: Creating AI-based platforms that integrate satellite imagery, IoT sensors, and data analytics to provide real-time insights for farmers. 2. Natural Language Processing (NLP) for Local Languages Focus: Developing NLP models that understand and process Ethiopian languages, such as Amharic, Oromo, and Tigrigna. Relevance: Language barriers can limit access to information and services. Research in NLP can enable the development of translation tools, voice assistants, and educational resources in local languages, improving accessibility and inclusivity. Innovation: Building large-scale language models for Ethiopian languages, enabling applications in areas like automated customer service, language learning, and digital literacy.	maintenance systems can analyze data from machinery and equipment to predict failures before they occur. This reduces downtime, extends equipment life, and lowers maintenance costs. Example: Implementing AI models that monitor industrial machines in factories to detect early signs of wear and tear, scheduling maintenance only when necessary, thereby optimizing operational efficiency. 2. Supply Chain Optimization Application: Retail, Logistics, Agriculture Solution: AI can optimize supply chains by predicting demand, managing inventory, and optimizing routes for delivery. This leads to reduced costs, improved efficiency, and better customer service. Example: Developing AI tools that forecast demand for agricultural products based on weather patterns, market trends, and historical data, helping farmers and retailers to manage inventory and pricing more effectively. 3. AI-Powered Customer Service Application: Banking, Telecommunications, E-commerce Solution: AI chatbots and virtual assistants can provide 24/7 customer support, handling routine inquiries, and resolving common issues without human intervention. This improves customer satisfaction and reduces operational costs. Example: Creating multilingual AI chatbots that understand and respond in Ethiopian languages, providing	

governance.		3. AI in Healthcare Focus: Applying AI to healthcare challenges such as disease diagnosis, personalized medicine, and healthcare delivery in remote areas. Relevance: Ethiopia faces significant healthcare challenges, including limited access to medical expertise and resources. AI can help bridge this gap by improving diagnostics, predicting disease outbreaks, and optimizing treatment plans. Innovation: Developing AI-driven diagnostic tools that leverage medical imaging, electronic health records, and wearable technology to improve early detection and treatment of diseases like malaria, tuberculosis, and non-communicable diseases. 4. Sustainable Development and Environmental Monitoring Focus: Using AI for environmental conservation, climate change mitigation, and sustainable resource management. Relevance: Ethiopia is vulnerable to environmental challenges such as deforestation, land degradation, and climate change. AI can contribute to	banking services, product support, or telecom troubleshooting to a diverse customer base. 4. Automated Fraud Detection Application: Banking, Insurance, E-commerce Solution: AI algorithms can detect fraudulent activities by analyzing patterns in financial transactions, insurance claims, or online purchases. These systems can flag suspicious behavior in real-time, preventing fraud before it occurs. Example: Implementing machine learning models in banks to analyze transaction data and detect anomalies that may indicate fraud, helping to secure financial systems and protect customers.	

		sustainable development by providing tools for monitoring environmental changes, managing natural resources, and supporting conservation efforts. Innovation: Creating AI models for real-time monitoring of ecosystems, predicting environmental risks, and optimizing resource use in sectors like water management and energy production. 5. Smart Cities and Urban Planning Focus: Developing AI solutions for urban planning, traffic management, and smart infrastructure. Relevance: As urbanization increases, cities in Ethiopia will need to become more efficient and sustainable. AI can help design smart cities by optimizing traffic flow, reducing energy consumption, and improving public services. Innovation: Implementing AI-driven systems for smart traffic management, energy-efficient building design, and urban analytics to enhance the quality of life in Ethiopian cities.		
Machine learning, critical thinking, socio-cultural awareness, best, research and banking(credit)	I have doing real projects when I was at 10 Academy, I recommend you	Gen AI, Autonomous systems, Machine learning	Smart chatbots (localized)	no

	to collaborate with them they are best.			
a) generela computer programing language, maths b) problem solving c)team-work,simplicity and decipline d)deployable and usable project artifact e) as it is emerging technology there will be many postions be available	i dont have	Data Science.... NLP, Geo-AI, public health improvement	Decision support systems, Predictive systems, Automation Systems	no i dont have